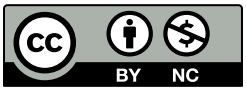
Introductory Mechanics Demonstration Documentation

This document was prepared as part of the PIO Instructional Development Grant Project: *Enhancing Classroom Demonstrations to Support Student Learning in Physics 2050 at Western Michigan University* by Siggi Rose and Jack Burkholder

The document was reviewed for accuracy by Shiva Agarwal. Although every effort has been made to ensure the accuracy of the material, some errors may remain.



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Vectors

What we need:

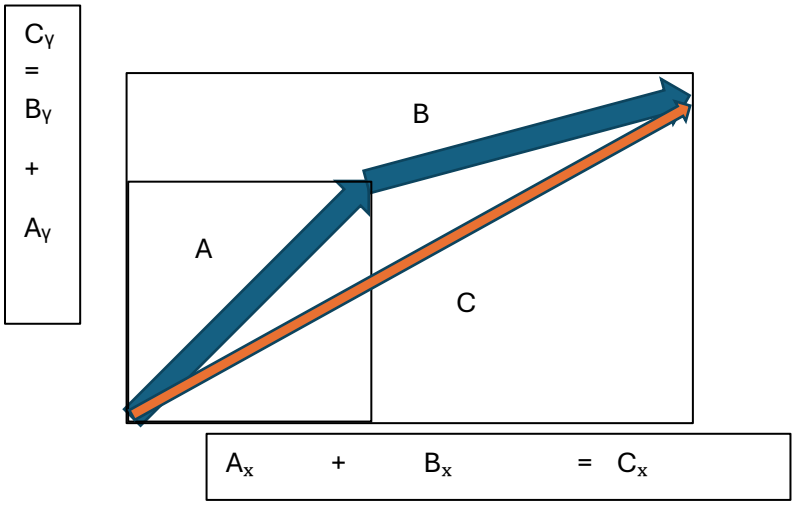
- 1 Whiteboard / blackboard
- 1 Marker / Chalk

What we do:

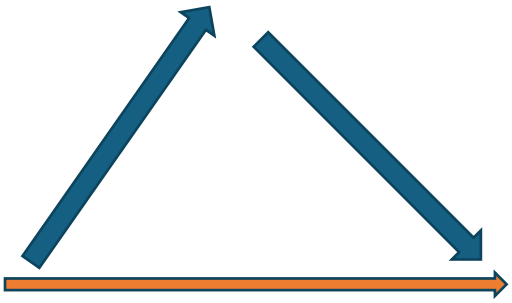
- Draw two vector arrows of the same length as so and note their identical lengths. Note that once you rotate one of these arrows, you are introducing a y-component.



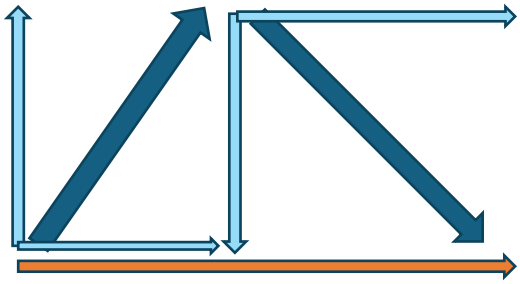
- Draw two more vector arrows of same length as so and note how they are constructive much like the first case, but with x and y components



- Draw the same length arrows at a slight angle and note that they are destructive in the y direction, but not in the x direction. Also note that the resultant vector is slightly shorter than in the first case.
    - Exaggerate angle so you can see the difference



- Split each vector arrow into its x and y components using arrows starting at the vector's origin.
    - Can do the same with the second drawing, but not necessary



- Finally, draw two vector arrows of the same length which essentially overlap and note how you end up back where you started.



- If class seems to be latching on, try the same concepts but with vectors of different lengths.

What concept we teach:

- Each vector has an x and a y component.
- When vectors are added together, they can be constructive, destructive, or somewhere in between. Some components can/will cancel out while others will continue to add/subtract and form a resultant.

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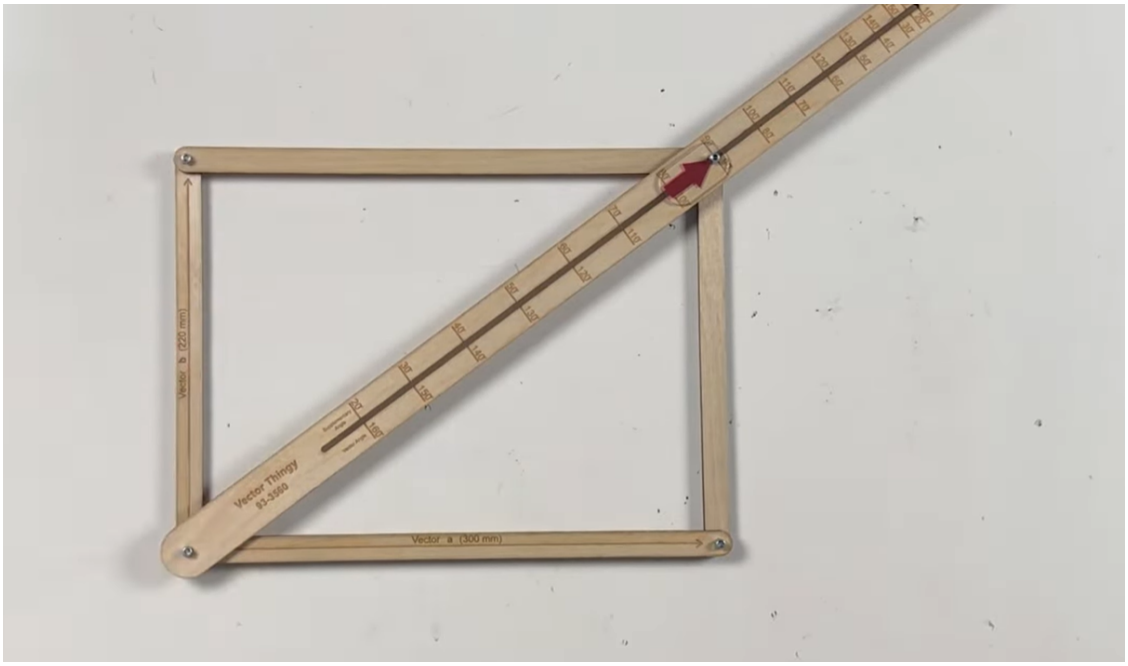
Vector Thingy

What we need:

- Vector thingy (can be found in the physics demo room with the rulers and meter sticks)

What we do:

- Use the vector thingy tool to help students visualize the resultant of two vectors.
    - Vector a is 300 mm and Vector b is 220 mm
- Start by setting the angle between Vectors a and b to 90 degrees.
  Vector Thingy 90 Degree Angle



  Note. From Arbor Scientific. (2024, March 13). *Vector Thingy Product Review* [Video]. Youtube. https://youtu.be/hkKOosY_WOU?si=ZyO0zP-ZrYr4HrI.

- Increase and decrease the angle to demonstrate how the magnitude of the resultant is affected by altering the angle between the two vectors.

Vector Thingy Altered Angle



Note. From Arbor Scientific. (2024, March 13). *Vector Thingy Product Review* [Video]. Youtube. https://youtu.be/hkKOosY_WOU?si=ZyO0zP-ZrYr4HrI.

- You can use the vector thingy to trace the vectors on the board to compare the altered angles and the resultants.

What concept we teach:

- As the angle between two vectors increases, the resultant decreases.
- As the angle between two vectors decreases, the resultant increases.

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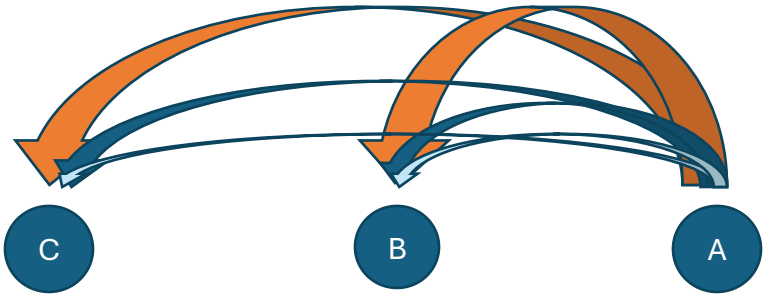
Projectile Motion

What we need:

- 3 Volunteers
- 1 Basketball

What we do:

- Ask volunteer 1 (shown as A in diagram) to take the basketball
- Volunteer 2 (shown as B in diagram) will stand in front of volunteer 1, volunteer 3 (shown as C in diagram) will stand behind volunteer 2
- Volunteer 1 will pass the ball to volunteer 2
- Volunteer 2 will hand the ball back to volunteer 1
- Volunteer 1 will now pass the ball to volunteer 3
- Discuss how the velocity or angle changed (or both depending on how the volunteer throws the ball to compensate for the larger distance
- Instruct volunteer 1 to throw the ball to volunteer 3 again, this time higher/lower than before.
- Discuss with the class how the only way to change how far the ball travels is to change the initial speed of the throw or the angle at which it is thrown.



What concept we teach:

- The same range can be achieved by altering either the initial velocity or the angle of the throw.
- To achieve a larger range, the initial velocity or the angle of throw must be increased. [Care should be taken as maximum range will be achieved at 45-degree angle of throw]

Range Equation

$$R = \frac{v_0^2 \sin(2\theta_0)}{g}$$

- The only variables that can contribute to a change in the range of a projectile are initial velocity and the angle of the throw.

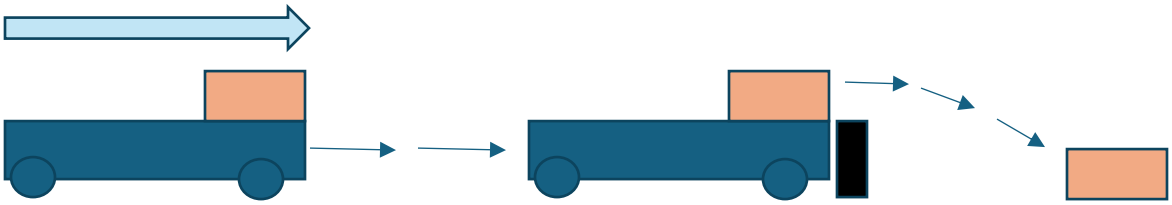
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Newton's First Law

What we need:

- 1 Low-friction cart
- 1 Meter stick
- 1 Blackboard eraser
- 2 Volunteers

What we do:



- Arrange the cart as shown on the left (the cart is dark blue, the eraser is orange)
- Position the meter stick standing up on edge and perpendicular to the motion of the cart, about a foot or two away. It should be the height of the cart, if not a little shorter.
    - Have a volunteer hold either side of the meter stick firmly. You may need one student to crouch down so the class can see
- Push the cart so that it makes contact with the meter stick, and the eraser continues its motion and falls to the table behind the meter stick.
    - Be careful not to push the cart too hard and move the eraser, but also not too soft that the eraser stays on the cart after contacting the meter stick.
- Discuss with the class how the eraser is able to continue its motion due to the lack of an opposing force.

    - While it is still being acted upon by the force of gravity, this force does not oppose motion as it acts in the -y direction.

What concept we teach:

- Newton's first law of motion states that an object continues its state of motion unless acted upon by non-zero net external force.

Helpful Notes:

- Do not mention friction or gravity as we found this did not help with analysis or explanation.

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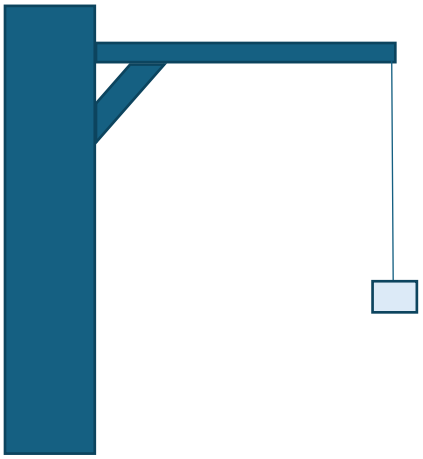
Circular Motion

What we need:

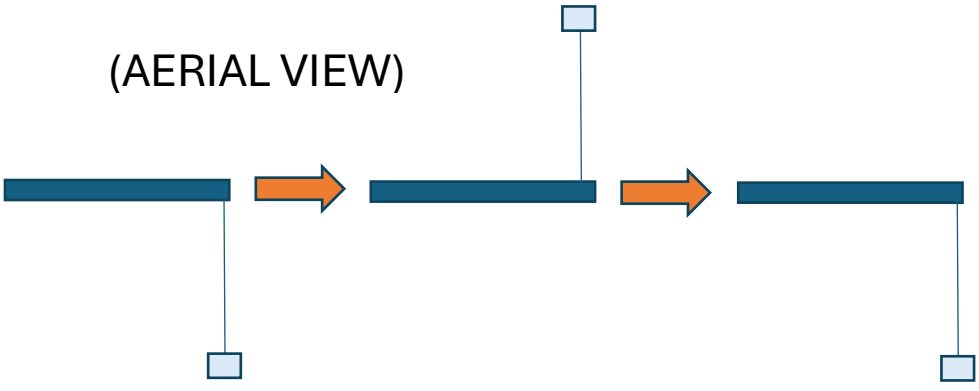
- 1 Metal Arm/Hook Mounted to Wall/Ceiling
- 1 String
- 1 Mass with hook on top

What we do:

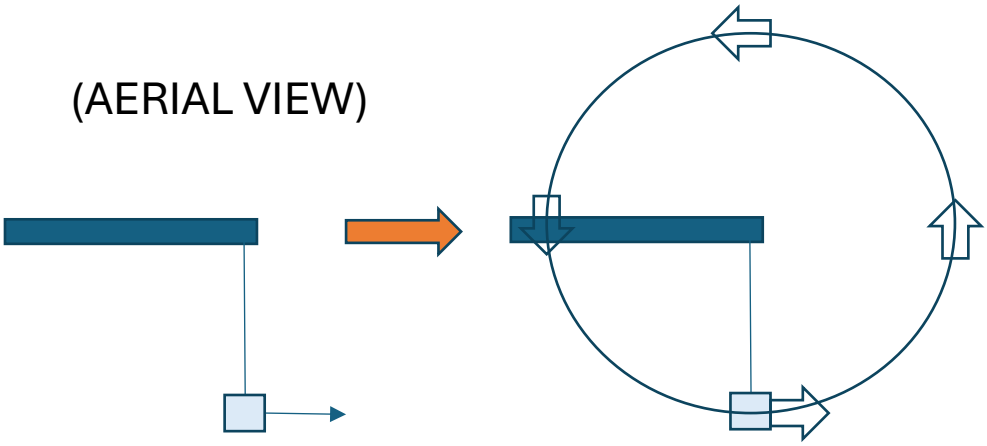
- Tie the string to the metal arm/hook and attach the hanging mass to the end of the string with a loop, like so

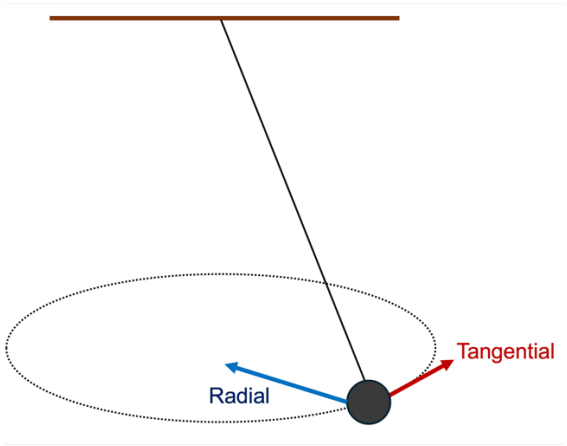


- Begin by pulling the string to the side so that the mass swings back and forth towards the mounted arm, like so



- Discuss with the class how the mass acts as a pendulum
- Now, grab the mass and stand directly under the mounted arm and face the class. Take a step or two to the right and throw the mass directly forward. It should follow a circular path like so.





- Discuss with the class that this is non-uniform circular motion due to energy lost by vibrations in the string and air resistance.

What concept we teach:

- The inward pull needed for circular motion is due to the radial component of the tension force. Without this force, the mass will keep moving in the tangential direction. If the mass is pushed or pulled in the radial direction, then it will only have back and forth motion.

Other ideas we had:

- Propel the hanging mass with a fan to simulate uniform circular motion. The fan will counteract the energy lost due to external forces and allow the mass to have uniform circular motion until the battery runs out. We theorized 3D modeling and printing something that would hold a small battery powered fan. Care must be taken for the center of mass of the combination as the fan should point in the plane of circular motion.

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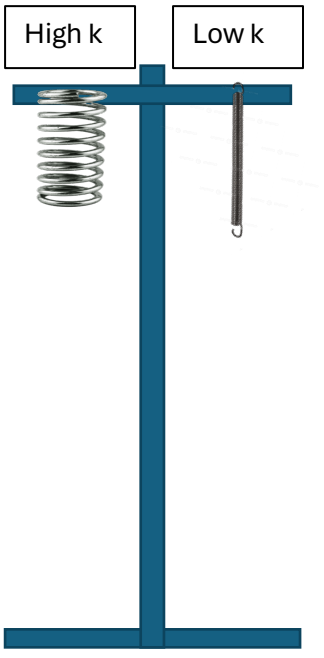
Hooke's Law

What we need:

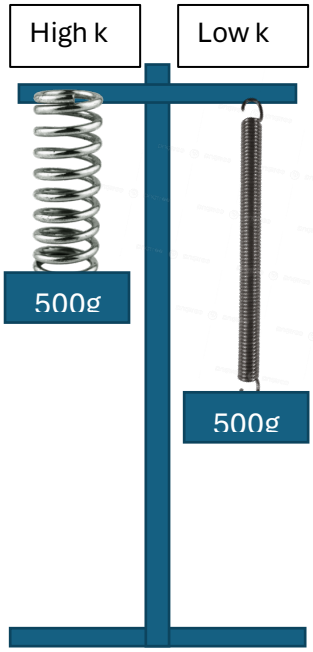
- 1 Spring with high spring constant
- 1 Spring with low spring constant
- 2 500g masses
- 2 1000g masses
- 1 Tall ring stand w/ 1 clamp

What we do:

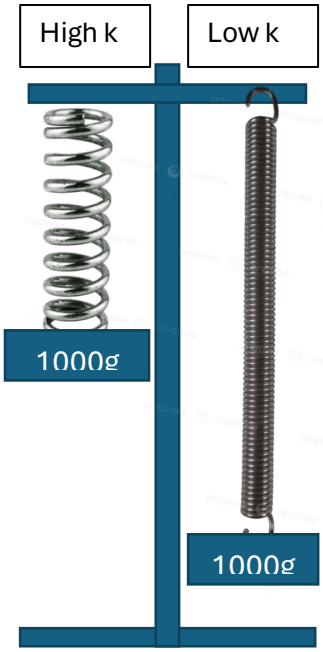
- Set the ring stand so that the clamp is near the top.
- Attach a spring to each side of the clamp so that they begin at the same vertical height.



- Attach a 500g mass to the end of each hanging spring and discuss with the class how despite the hanging mass being the same in both cases, the variation in spring constants leads to a variation in distance from its resting position.



- Now, remove the 500g masses and attach a 1000g mass to each spring. Discuss with the class how the force of gravity exerted by the hanging mass is greater, but the spring constant remains the same so the spring will stretch a greater distance as a result.



- If students seem confused, it might help to break down the units of the spring constant. The value in N / m indicates how many newtons are required to stretch the spring by one meter; a larger value means it takes more force to stretch the spring per meter, and a smaller value takes less force to stretch the spring per meter.

What concept we teach:

- Spring constant is a measure of the force needed to stretch a spring one meter.
- In this case, you can directly calculate the spring constant by equating the spring force to the gravitational force

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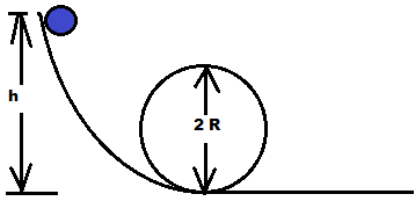
Conservation of Energy: Loop-de-Loop

What we need:

- 1 Loop-de-loop
- 1 Rubber Ball
- 1 Volunteer

What we do:

- Set up the Loop-de-loop on the front table and draw the situation on the blackboard. Ask the class to guess where they think you will need to drop the ball from so that it will complete the full circular motion.



- Ask a volunteer to stand at the end of the table to catch the ball, and drop the ball from the spot the class decides on.
- Discuss with the class how the height at which to drop the ball can be calculated using the conservation of energy.

$$KE_i + U_i = KE_f + U_f$$

$$0 + mgh = \tfrac{1}{2}mv_f^2 + mg(2R) \qquad (v_f^2 = gR)$$

$$h = \tfrac{1}{2}R + 2R \qquad h = 2.5R$$

- Measure the diameter of the loop, halve it, and demonstrate dropping the ball at the calculated height. Explain that it will not complete the full circle of motion due to energy lost due to friction and heat.
- Drop the ball from slightly higher up on the ramp to demonstrate that the ball will complete a full rotation.

What concept we teach:

- In loop-de-loop problems, to complete the full circular motion, the ball must be dropped from a height that is 2.5x the radius of the circle.
- Extra height is needed to overcome the real-world values that are not being accounted for in our calculation.

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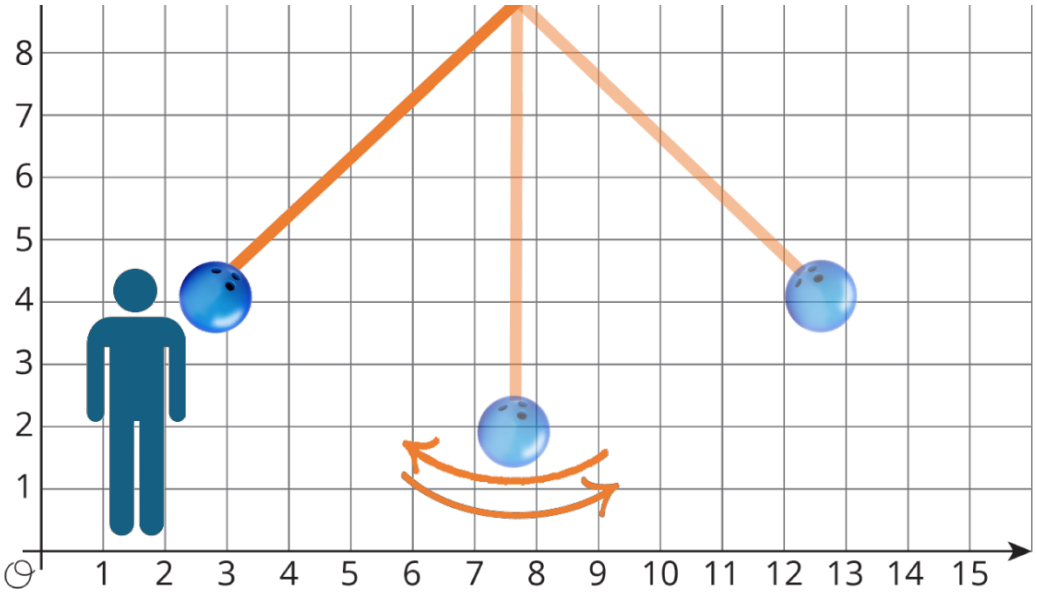
Conservation of Energy: Bowling Ball

What we need:

- 1 bowling ball
- 1 long rope/cord
- 1 volunteer

What we do:

- Attach the rope to the ceiling and then attach the bowling ball to the rope.
- Ask a volunteer to stand to the side and hold the bowling ball out in front of their face and then ask them to release the bowling ball from rest.
    - NOTE: Make sure the volunteer just releases the ball and does not push or throw the ball. Also ensure that the volunteer remains in their initial position.
- The bowling ball should swing across the room and then swing back towards the volunteer and stop at or around the same position that it was initially released.



- Discuss with the class why the bowling ball cannot hit the volunteer when it swings back towards them.
    - It may also be helpful to discuss the external, nonconservative forces that cause a slight loss of mechanical energy of the bowling ball.

What concept we teach:

- The bowling ball starts with a certain amount of gravitational potential energy while it is at rest in the air, which is transferred into kinetic energy as it swings and then once again is transferred into gravitational potential energy as it stops.
- The total amount of potential energy and kinetic energy will always remain constant, if no other forces are acting on the bowling ball. The bowling ball will not have enough energy to swing higher than its initial position.
- The change of the total energy of the bowling ball is equal to 0.

Note: It may be a good idea for the instructor to perform this demo by themselves and then let any volunteer to try this under the instructor's supervision.

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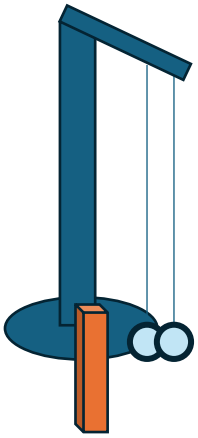
Momentum: Ball & Block Collisions

What we need:

- 2 balls of the same mass: one must be bouncy and the other should not bounce (available in physics demo room conservation of momentum box [purchased through Arbor Scientific])
- 1 pendulum stand
- String
- 1 wooden or plastic block

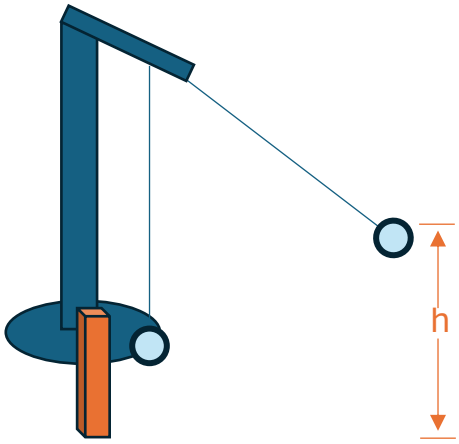
What we do:

- Show the class that one ball bounces and the other does not.
- Tie both balls so they each hang from the stand at the same height.
- Place the wooden block upright at the base of the stand.



- Ask the class which ball is most likely to knock over the wooden block.

- Lift the non-bouncy ball up to a certain height and then release it from rest. It should swing back down and collide with the wooden block without bouncing off the block or knocking it over.



- Repeat the previous step with the bouncy ball by lifting it and releasing it from the same height as the other ball. (Note: both balls must start with the same gravitational potential energy.) The ball should swing down and collide with the wooden block, but this time the ball will bounce off the block after the collision and the block should be knocked over.

- Watch this video from 2:34-3:20 to see an example of the demonstration:



https://youtu.be/_DA9_Hd372I?si=dS5fzEJb3rXuCkNg

- Discuss conservation of momentum and the difference between the two collisions with the class. Also discuss why the final momentum of the block is different depending on the collision.

What concept we teach:

- Total momentum is conserved in all collisions.
- Kinetic energy is conserved in elastic collisions, but not inelastic collisions.
- In elastic collisions, more momentum is transferred than in inelastic collisions.

Note:

In our case, collision by rolling the balls on a ramp worked better. This video may be helpful: https://youtu.be/oTuibz11FFU?si=85h3Sk5ajPaNMbm0

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Momentum: Stacked Balls

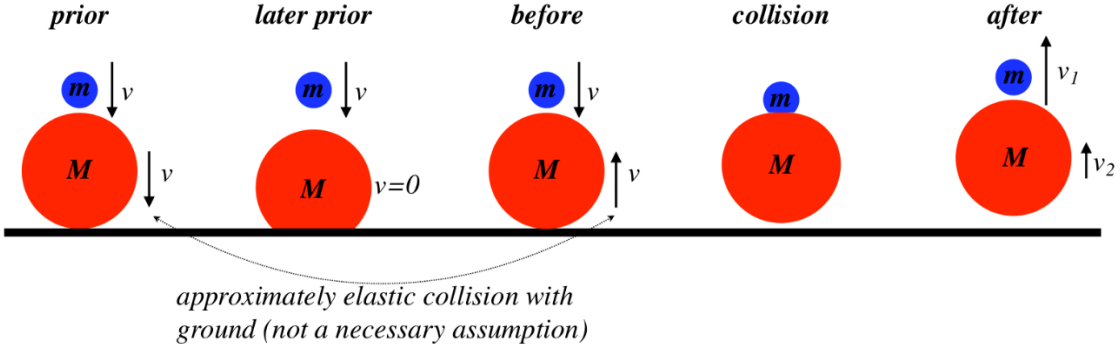
What we need:

- At least 2 different sized bouncy balls (ex: a basketball and tennis ball)

What we do:

- Drop each ball separately from the same height to demonstrate to the class how high they bounce off the floor.
- Stack the balls on top of each other with the smallest ball on the top. Make sure all the balls are centered. Then, release all the balls at once from rest. After hitting the floor, the momentum of the system should all transfer to the smallest ball on top, which will make it rebound to a much higher position than the height it was dropped.
    - Note: Increasing the number of stacked balls will increase the height that the small ball on top will be propelled

Stacked Balls



Note. From LibreTexts. (2024). 4.6 Problem Solving. In T. Weideman, *UCD: Physics 9HA – Classical Mechanics.* Davis: University of California, Davis.

- Discuss with the class how the momentum of the system of balls is transferred through to the ball on top as they collide with the floor and then collide with each other.

What concept we teach:

- Momentum is transferred through objects in collisions.
- Total momentum is conserved in all collisions.
- Mechanical energy is conserved in elastic collisions.

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Rolling Motion of a Rigid Object

What we need:

- 1 ramp
- 2 solid balls with different radii
- 1 hollow ball
- 1 solid cylinder
- 1 hollow cylinder

What we do:

- Take the equation for the kinetic energy of a rolling object and the equation for conservation of mechanical energy to find the velocity of the center of mass of the rolling objects. Notice how mass and radius cancel out and do not affect the velocity.

$$KE_{rolling}=KE_{tangential}+KE_{rotational}=\frac{1}{2}Mv_{cm}^{2}+\frac{1}{2}I\left(\frac{Vcm}{R}\right)^{2} \qquad \Delta KE+\Delta U=0$$

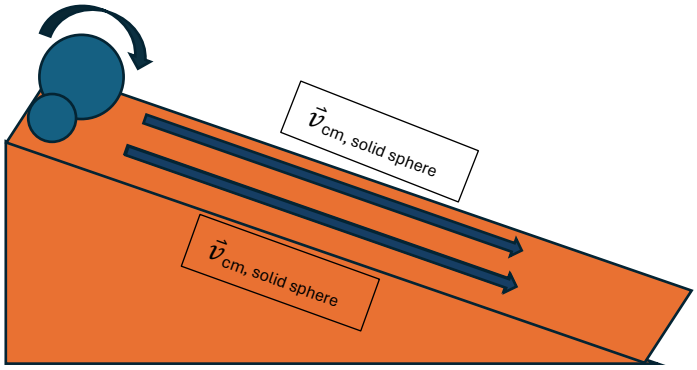
$$V_{cm}=\left(\frac{2gh}{1+\frac{I}{MR^{2}}}\right)^{1/2}$$

$$V_{cm,\ solid\ sphere}=\left(\frac{2gh}{1+\frac{2}{5}}\right)^{1/2} \quad V_{cm,\ spherical\ shell}=\left(\frac{2gh}{1+\frac{2}{3}}\right)^{1/2}$$

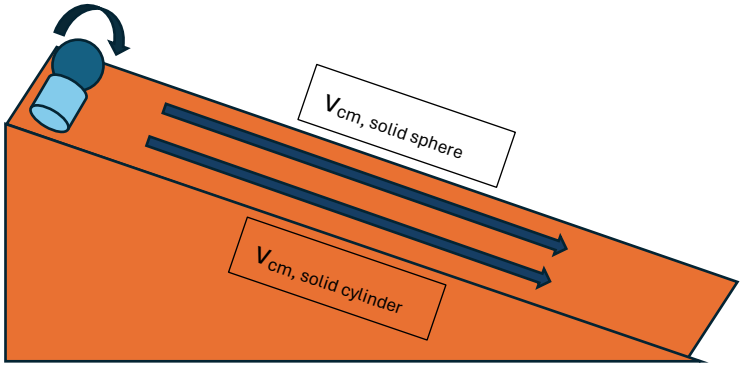
$$V_{cm,\ cylindrical\ shell}=\left(\frac{2gh}{1+1}\right)^{1/2} \quad V_{cm,\ solid\ cylinder}=\left(\frac{2gh}{1+\frac{1}{2}}\right)^{1/2}$$

- Ask the class which object would roll faster based on these equations.

- Demonstrate what happens when two solid spheres with different radii and masses roll down the ramp when released from the same height at the same time.



- Demonstrate what happens when a solid sphere and a solid cylinder roll down the ramp when released from the same height at the same time.



- Continue demonstrating with different combinations of the objects and compare their velocities.

What concept we teach:

- Gravitational potential energy translates into rolling kinetic energy.
- Moment of inertia of objects changes with different shapes. Objects with a lower moment of inertia roll faster.
- Velocity of center of mass of an object with simple geometry depends on its shape and not on the mass and size.
- Conservation of mechanical energy for sphere-Earth system.

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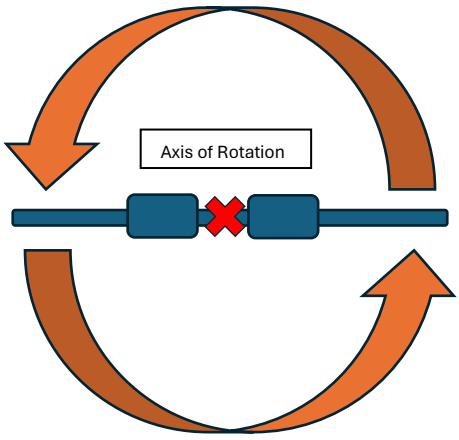
Rotational Motion: Turning Rod

What we need:

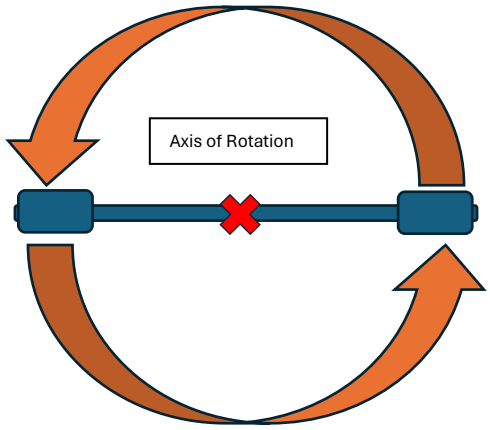
- 1 rod with 2 weights attached (can be borrowed from PHYS 1140 lab)
- 1 volunteer

What we do:

- Ask a volunteer to hold the rod by its center with the two weights close to the center. Then ask them to rotate the rod.



- Now move the weights to each end of the rod and ask the volunteer to try to rotate the rod once again from its center. It should be harder for them to rotate compared to when the weights were closer to the center.



- Discuss how weight distribution affects rotational inertia, and the amount of energy required to rotate the rod.

What concept we teach:

- Mass distribution affects the rotational inertia of the system.
- According to the equation for rotational kinetic energy ($KE_{rot}=\frac{1}{2}I\omega^2$), increasing the rotational inertia also increases the energy needed to rotate the rod with the same angular velocity.
- Changing the rotational inertia also affects how much torque is required to produce the same angular acceleration ($\tau=I\alpha$).

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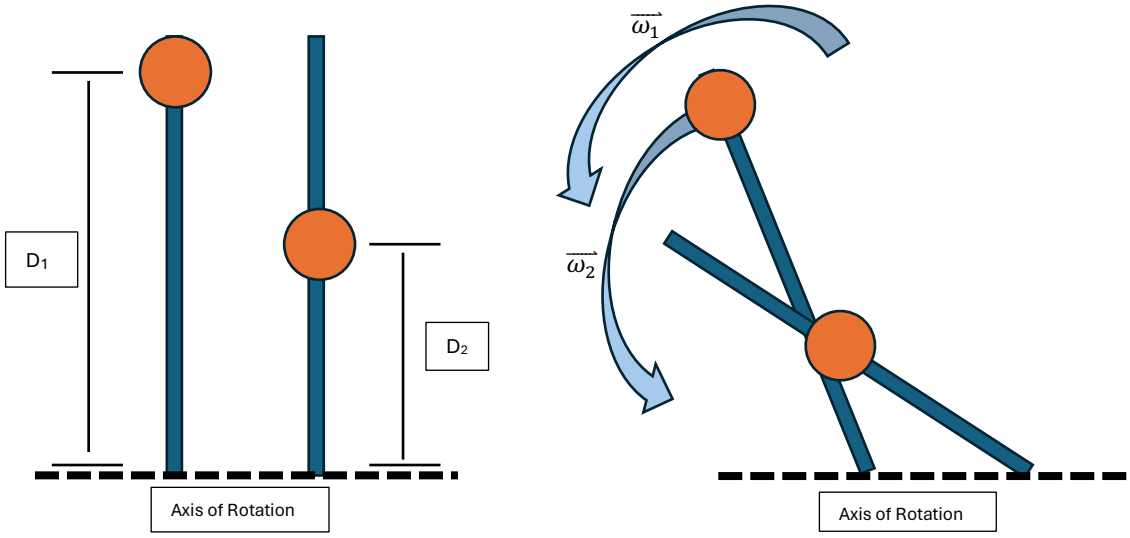
Rotational Motion: Falling Rods

What we need:

- 2 sturdy sticks/rods/dowels of same mass and length
- 2 masses of the same shape and mass (we used 2 tennis balls, but you might want to use something a little heavier)

What we do:

- Attach one mass to the end of one of the sticks and attach the other mass to the center of the other stick.
- Hold both sticks upright and then release them from rest. They should pivot about one end and fall due to gravity.



- Discuss with the class why one stick falls faster than the other.

What concept we teach:

- Mass distribution affects the rotational inertia of the system.
- Rotational inertia affects angular velocity. A system with less rotational inertia needs less energy to move and therefore has a greater angular velocity.
- Gravitational potential energy translates into rotational kinetic energy.
- Although the stick with the mass attached to its upper end starts with more gravitational potential energy, it still has a lower angular velocity due to it having a greater rotational inertia.

Note: We had difficulty getting consistent results while testing this demonstration. This is most likely because all we had to use were two metal yard sticks, which were flimsy.

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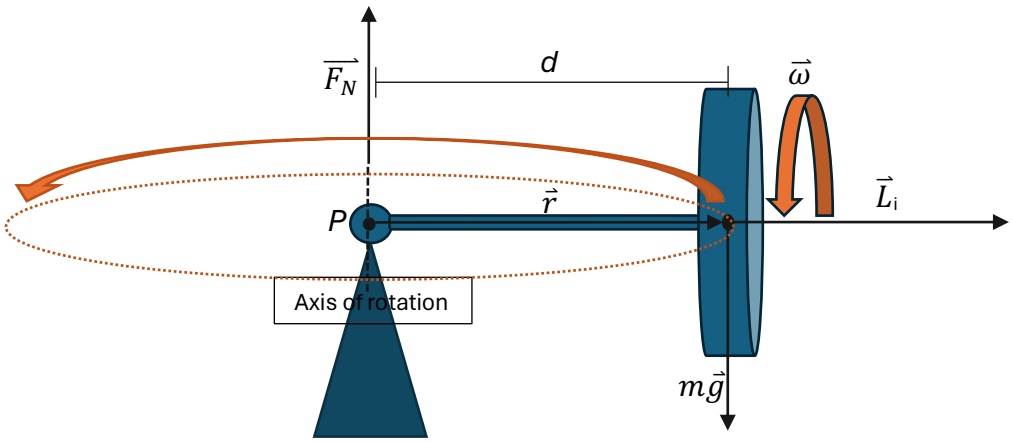
Angular Momentum: Gyroscope

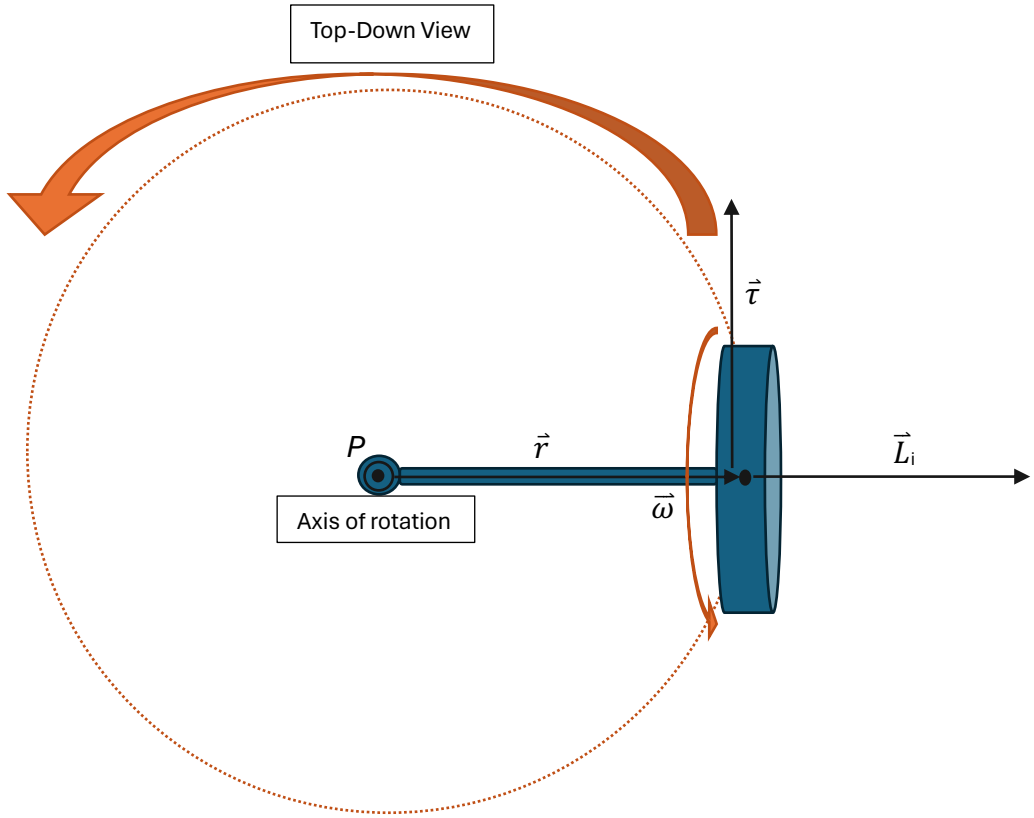
What we need:

- 1 gyroscope
- 1 pedestal to support the axle of the gyroscope
- String or handle to spin gyroscope

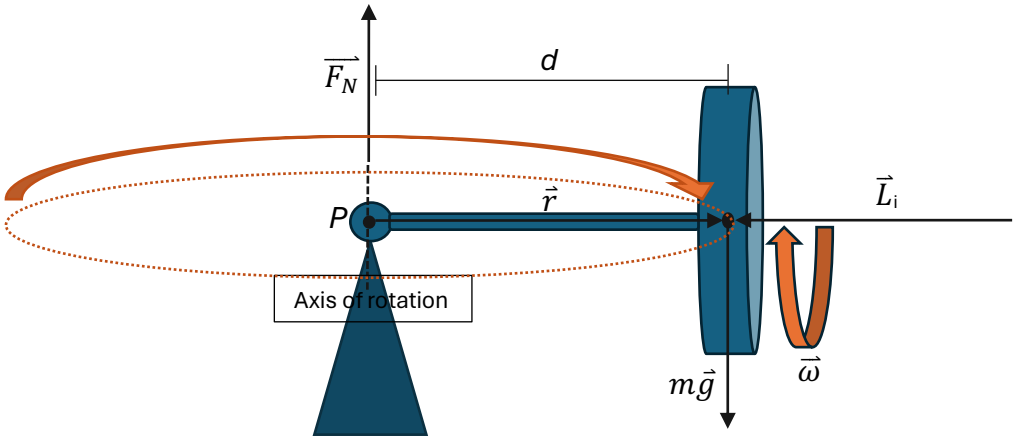
What we do:

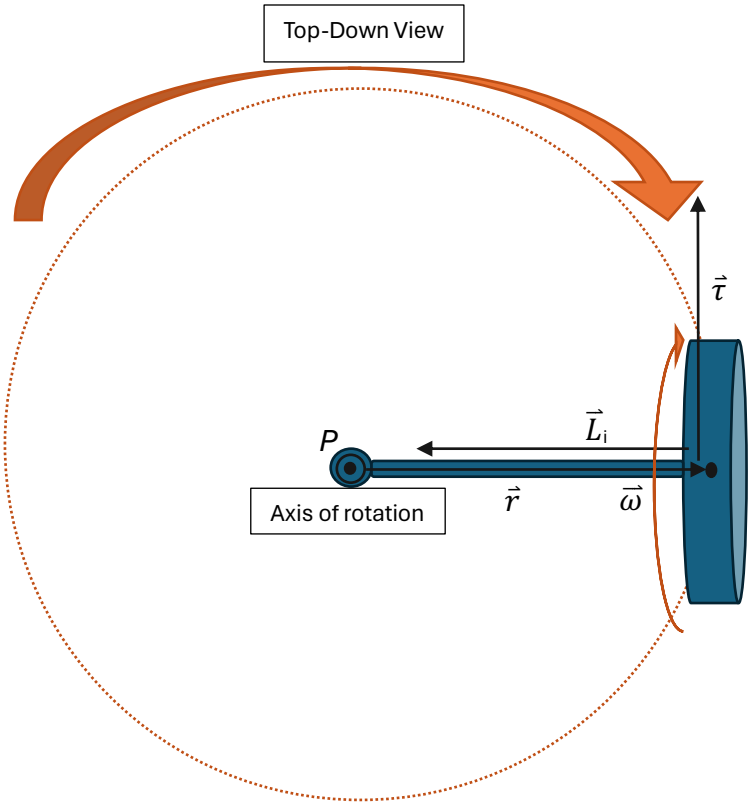
- To spin the wheel of the gyroscope, either wrap a string around the axle or insert the T-handle, then pull in a quick motion.
- Line up the end of the axle of the gyroscope with the pedestal and then release the gyroscope. If the wheel of the gyroscope spins in the clockwise direction, the gyroscope will precess in the clockwise direction. Similarly, the gyroscope will precess in the counterclockwise direction if the gyroscope wheel spins in the counterclockwise direction.
    - **Counterclockwise:**




Top-Down View
$\vec{\tau}$
P
$\vec{r}$
$\vec{L}_i$
$\vec{\omega}$
Axis of rotation

- **Clockwise:**


$\overrightarrow{F_N}$
d
P
$\vec{r}$
$\vec{L}_i$
Axis of rotation
$m\vec{g}$
$\vec{\omega}$



- Draw a diagram of the gyroscope and explain the dynamics of angular momentum and torque and how they cause the gyroscope orbit around the pedestal without falling.

What concept we teach:

- The force of gravity creates an external torque, but the angular momentum from the gyroscope spinning prevents it from falling.
- The initial direction of the angular momentum depends on the direction that the wheel of the gyroscope is spinning.
- Torque causes the angular momentum to change direction, thus causing the gyroscope to precess around the pedestal.
- The directions of torque and angular momentum remain perpendicular to each other.

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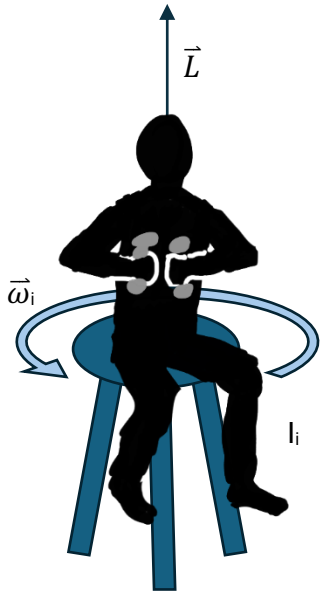
Angular Momentum: Rotating Stool

What we need:

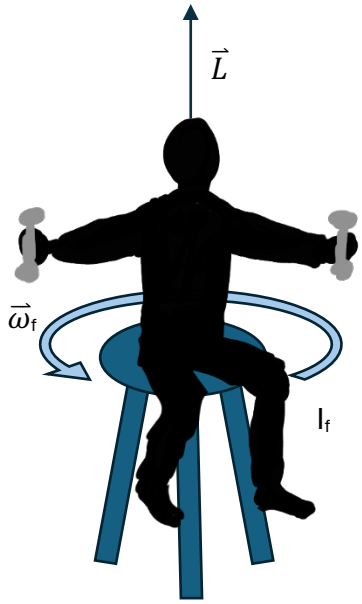
- 1 stool/chair that spins
- 2 dumbbells of the same mass
- 1 volunteer

What we do:

- Ask a volunteer to sit on the stool/chair while holding the dumbbells close to them in each hand. Ask the volunteer to spin the stool.



- Ask the volunteer to extend their arms out and hold the dumbbells further from their center of mass. Their angular velocity should decrease.



- Now ask the volunteer to bring the dumbbells back closer to their center of mass. Their angular velocity should increase to a magnitude similar to when they started spinning.
- The volunteer may keep changing the distance of the dumbbells to their center and observe the changes in their angular velocity.

What concept we teach:

- Conservation of angular momentum.
- Mass distribution affects the rotational inertia of the system.
- When angular momentum does not change, then the moment of inertia and angular velocity are inversely proportional to each other ($L=I\omega$).

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Universal Gravitation

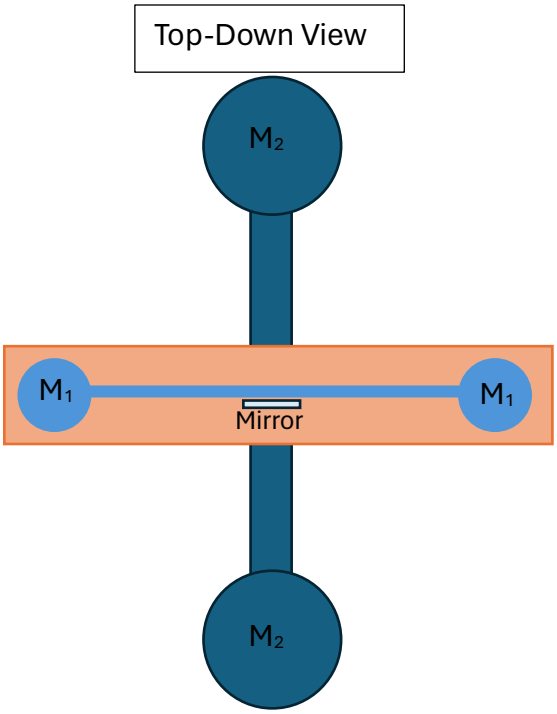
What we need:

- Cavendish Balance
- 2 large lead spheres
- 1 laser
- Chalkboard or white board

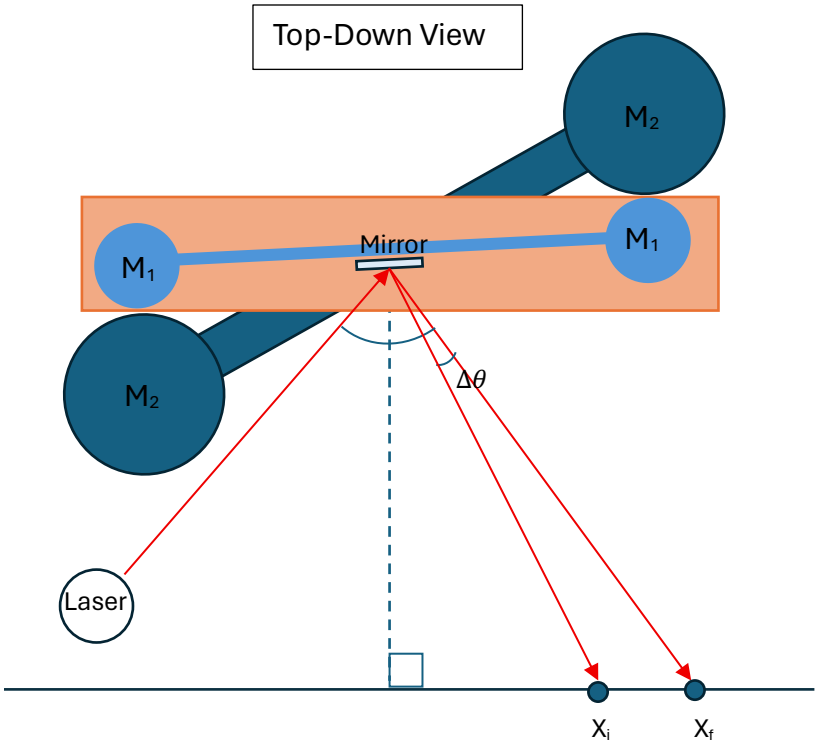
Note: We had a lot of trouble with this demonstration, as the original torsion band was missing from the Cavendish apparatus available in physics demo room and was expensive to replace. We attempted to use a cheaper wire but were unsuccessful in getting results. For the future, thread may be a good alternative and would be worth testing out. However, if the cavendish balance is being used to solve for the gravitational constant, then the torsion band will probably be required.

What we do:

- Carefully place the balance on a stable surface where there are minimal vibrations or disturbances.
- Make sure that the balance is as level as possible and adjust the knobs at the bottom of the stand if needed.
- Place the larger lead spheres on the lower support arm at a position that is away from the smaller spheres.



- Set up the laser from a distance so that it is level with and pointing at the small mirror so that it is reflected onto the board.
- Allow the system to come to rest into a stable equilibrium position, then mark it on the board.
    - Note: This may take a while.
- Quickly (but carefully) swivel the lower supporting arm so that the large spheres are touching or nearly touching the glass encasing the smaller spheres.



- Mark the position of the laser on the board every 30 seconds as the system oscillates to show to the class that there is acceleration.
    - Note: There may not be enough time in class to see a full oscillation, but the students will still be able to see that the system is accelerating.
- Let the system come to a rest once again and then mark the new equilibrium position.
- For more detailed information about the balance, refer to the instruction sheet here: https://labdocs.leylab.de/doc/en/GA/PDF/3/332/332101de.pdf?hash=alShTnKD
- The Cavendish balance can be used to determine the gravitational constant, *G*, using the equations for Newton's law of universal gravitation

($F = G\frac{m_1 m_2}{r^2}$) and torque ($\tau = k \cdot \Delta\theta$) and the torsion constant, *k*, which can be found using the period of oscillation equation for a torsion balance ($T = 2\pi\sqrt{\frac{I}{k}} \rightarrow k = \frac{4\pi^2 \cdot I}{T^2}$). Torque is related to force: $\tau = Fd$.

Solve for *G*:

$$k \cdot \Delta\theta = G\frac{m_1 m_2}{r^2}d \rightarrow G = \frac{k \cdot \Delta\theta \cdot r^2}{d \cdot m_1 m_2}$$

- This document details different methods of determining *G*: https://labdocs.leylab.de/doc/en/EXP/P/P1/PDF/P1131_e.pdf?hash=DjEkA+4D

What concept we teach:

- Newton's law of universal gravitation

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Resources

- Arbor Scientific: Online shop specifically for physics equipment, supplies, and resources. Link: https://www.arborsci.com/
- Leybold: Online shop with a wide range of equipment and spare parts that can be used for physics demonstrations. Link: https://www.leybold-shop.com/

Works Cited

Arbor Scientific. (n.d.). Arbor Scientific: Physics and Chemistry Science Lab Equipment. Retrieved from Arbor Scientific: https://www.arborsci.com/

Arbor Scientific. (2019, March 29). 7 Energy and Momentum Demos That Will Engage Your Students [Video]. Youtube. https://youtu.be/_DA9_Hd372I?si=Cn1PhyQToqTaUbF4

Arbor Scientific. (2024, March 13). Vector Thingy Product Review [Video]. Youtube. https://youtu.be/hkKOosY_WOU?si=ZyO0zP-ZrYr4HrI

LD Didactic GmbH. (n.d.). Determining the gravitational constant with the gravitation torsion balance after Cavendish. Federal Republic of Germany. Retrieved from https://labdocs.leylab.de/doc/en/EXP/P/P1/PDF/P1131_e.pdf?hash=DjEkA+4D

LD Didactic Group. (n.d.). LD DIDACTIC | Natural Sciences, Vocational Training, Physics ... Retrieved from Leybold: https://www.leybold-shop.com/

Leybold Didactic GmbH. (n.d.). Instruction Sheet Gravitation Torsion Balance. Federal Republic of Germany. Retrieved from https://labdocs.leylab.de/doc/en/GA/PDF/3/332/332101de.pdf?hash=alShTnKD

LibreTexts. (2024). 4.6 Problem Solving. In T. Weideman, UCD: Physics 9HA – Classical Mechanics. Davis: University of California, Davis.

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AM Surveys

Poll results

slido

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- 9am Demo 1: Projectile Motion
- 9am Demo 2 PRE: Laws of Motion
- 9am Demo 2 POST: Laws of Motion
- 9am Demo PRE: Circular Motion
- 9am Demo 3 POST: Circular Motion
- 9am Demo 4 PRE: Loop-de-Loop
- 9am Demo 4 POST: Loop-de-Loop
- 9am Demo 5 PRE: Pendulum
- 9am Demo 5 POST: Pendulum
- 9am Demo 6 PRE: Collisions
- 9am Demo 6 POST: Collisions
- 9am Demo 7 PRE: Rolling Down a Ramp
- 9am Demo 7 POST: Rolling Down a Ramp
- 9am Demo 8 PRE: Dumbbell Torque

slido

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- 9am Demo 8 POST: Dumbbell Torque

9am Demo 1: Projectile Motion (1/5)

0 4 2

In this demo, three students are arranged in a straight line. One student throws a basketball in an arc to the first (closer) student, and then to the second (farther) student.

(1/2)

The student could adjust either the initial velocity or the angle of the basketball to complete the second pass.

95 %

The student must only adjust the angle of the basketball to complete the second pass.

2 %

The student must only adjust the initial velocity of the basketball to complete the second pass.

2 %

9am Demo 1: Projectile Motion (1/5)

0 4 2

In this demo, three students are arranged in a straight line. One student throws a basketball in an arc to the first (closer) student, and then to the second (farther) student.

(2/2)

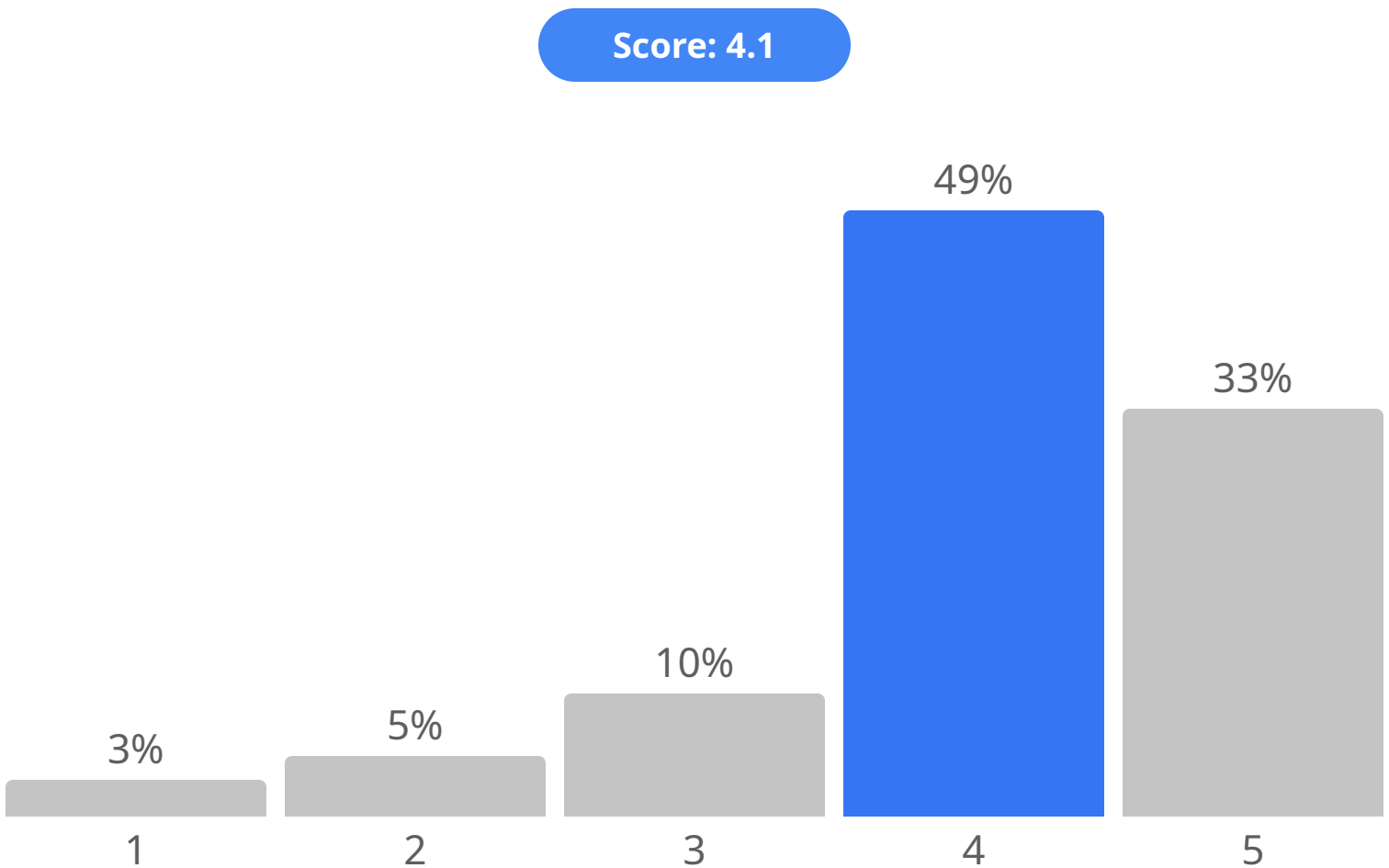
The student needs to adjust something else to complete the second pass.

0 %

9am Demo 1: Projectile Motion (2/5)

0 3 9

On a scale from 1-5, rate your confidence in your answer.



9am Demo 1: Projectile Motion (3/5)

0 4 2

In this demo, three students are arranged in a straight line. One student throws a basketball in an arc to the first (closer) student, and then to the second (farther) student.
(1/2)

The kinematic equations cannot be used to analyze this problem, because acceleration will be constant.

24 %

The kinematic equations cannot be used to analyze this problem, because acceleration will not be constant.

5 %

The kinematic equations can be used to analyze this problem, because acceleration will be constant.

62 %

9am Demo 1: Projectile Motion (3/5)

0 4 2

In this demo, three students are arranged in a straight line. One student throws a basketball in an arc to the first (closer) student, and then to the second (farther) student.

(2/2)

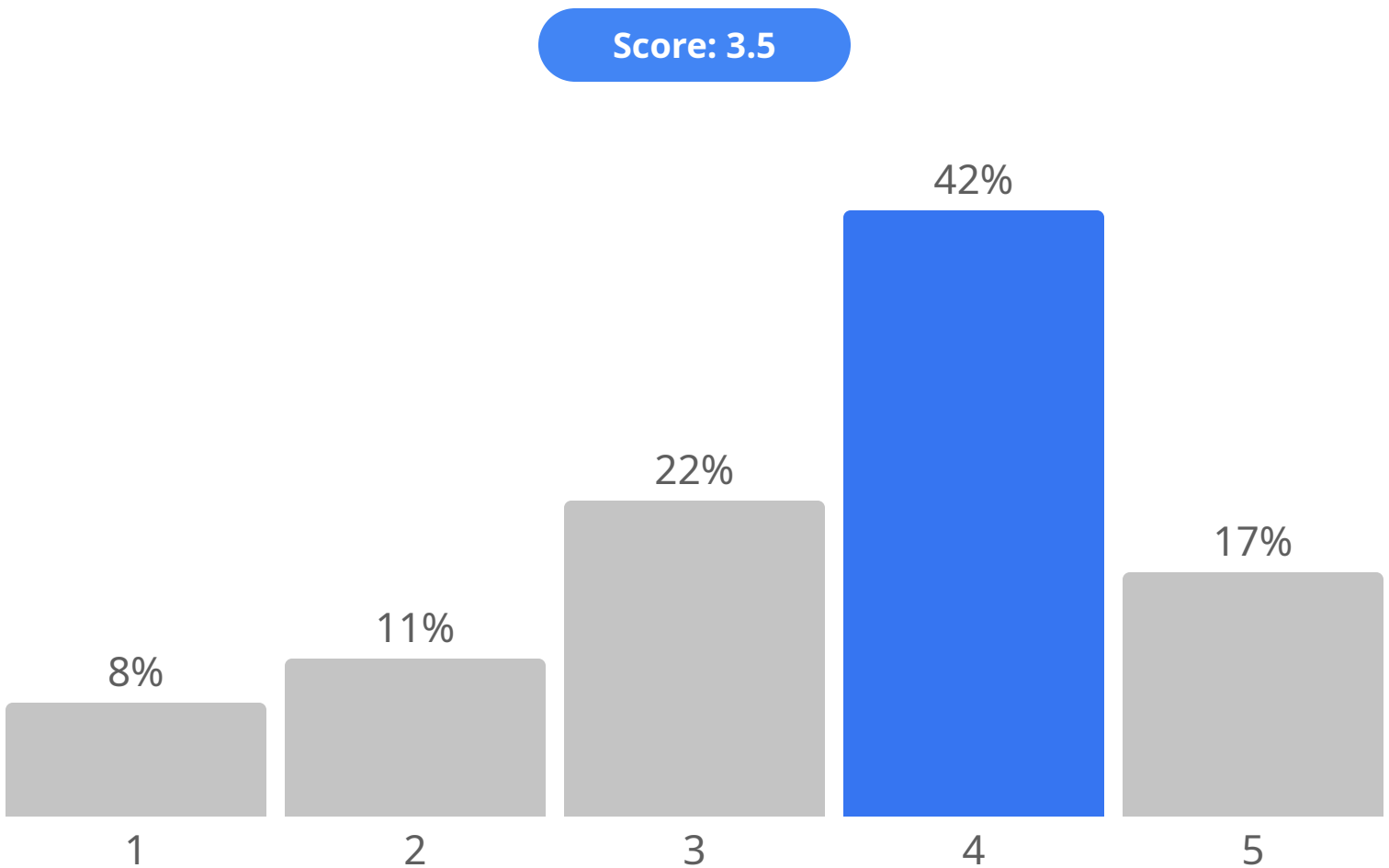
The kinematic equations can be used to analyze this problem, because acceleration will not be constant.

10 %

9am Demo 1: Projectile Motion (4/5)

0 3 6

On a scale from 1-5, rate your confidence in your answer.



9am Demo 1: Projectile Motion (5/5)

0 4 2

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

29 %

Demonstration setup

2 %

Demonstration outcome

57 %

Physics equations

7 %

Something else

5 %

9am Demo 2 PRE: Laws of Motion (1/2)

0 3 9

In this demo, two objects of comparable mass are attached by a string, which runs through a pulley. One object sits atop a table, while the other is suspended in the air. The suspended object will be released from rest.

Only the hanging mass will move.

3 %

The hanging mass starts falling, and after a few seconds the other mass moves.

5 %

The two masses both move, but the hanging mass will move faster.

8 %

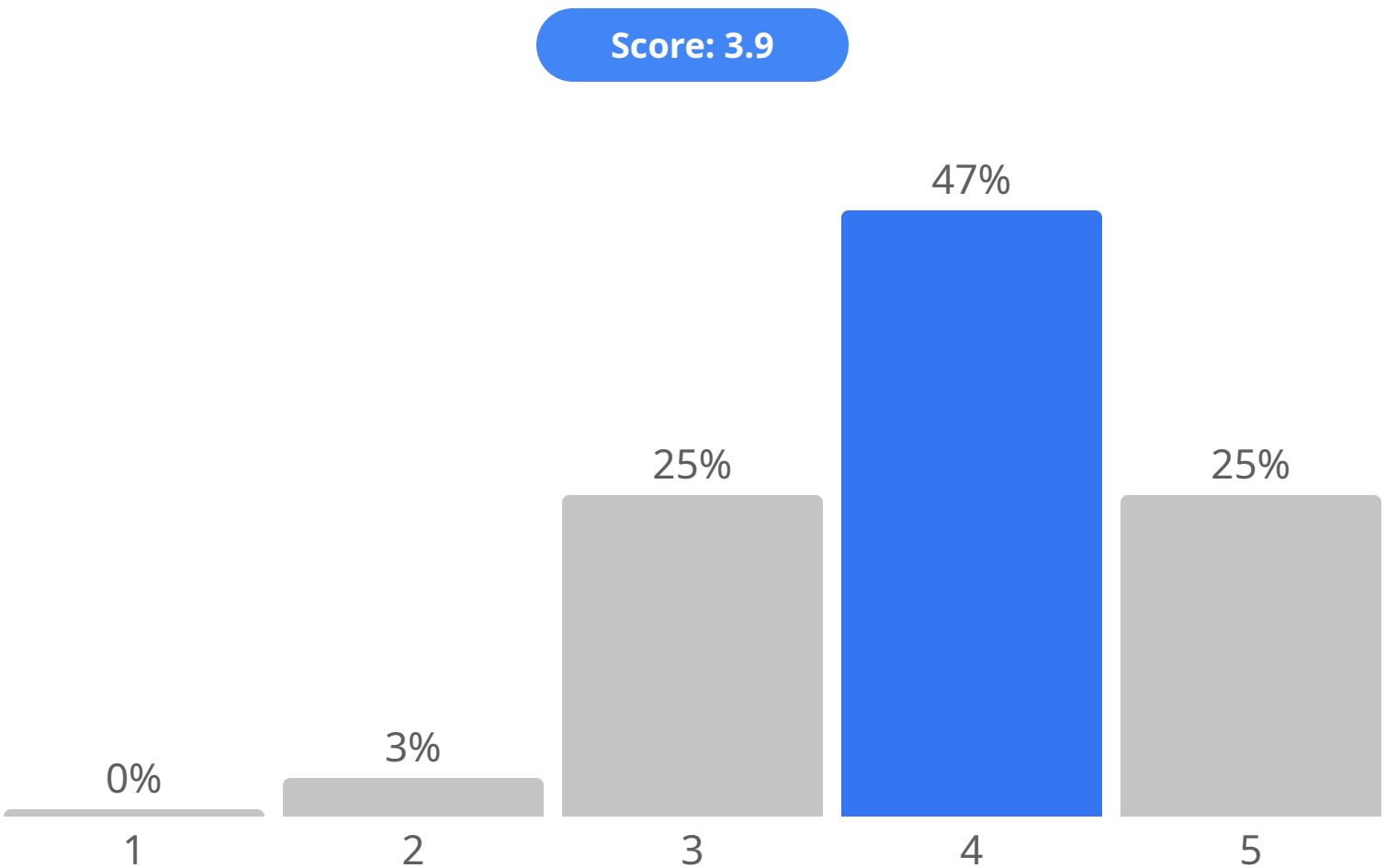
The two masses will move with the same speed.

85 %

9am Demo 2 PRE: Laws of Motion (2/2)

0 3 2

On a scale from 1-5, rate your confidence in your answer.



9am Demo 2 POST: Laws of Motion (1/3)

0 3 8

In this demo, two objects of comparable mass are attached by a string, which runs through a pulley. One object sits atop a table, while the other is suspended in the air. The suspended object will be released from rest.
(1/2)

The hanging mass will not accelerate at all, it will fall with constant speed.

11 %

The hanging mass will fall with acceleration smaller than 9.8 m/s^2, due to the object on the table.

82 %

The hanging mass will fall with acceleration 9.8 m/s^2, because the mass on the table does not affect acceleration.

8 %

9am Demo 2 POST: Laws of Motion (1/3)

0 3 8

In this demo, two objects of comparable mass are attached by a string, which runs through a pulley. One object sits atop a table, while the other is suspended in the air. The suspended object will be released from rest.
(2/2)

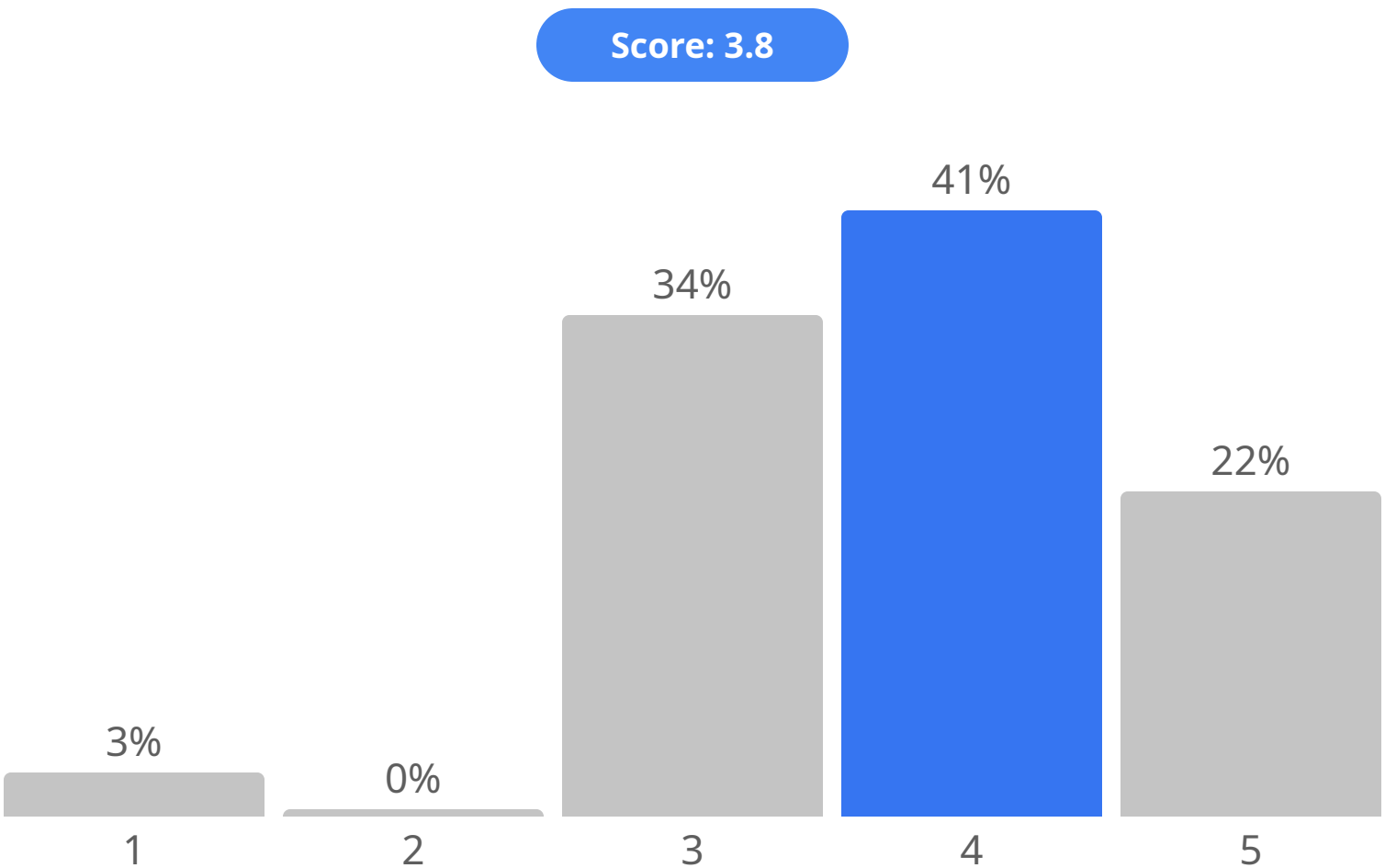
The hanging mass will fall with acceleration greater than 9.8 m/s^2, due to the object on the table.

0 %

9am Demo 2 POST: Laws of Motion (2/3)

0 3 2

On a scale from 1-5, rate your confidence in your answer.

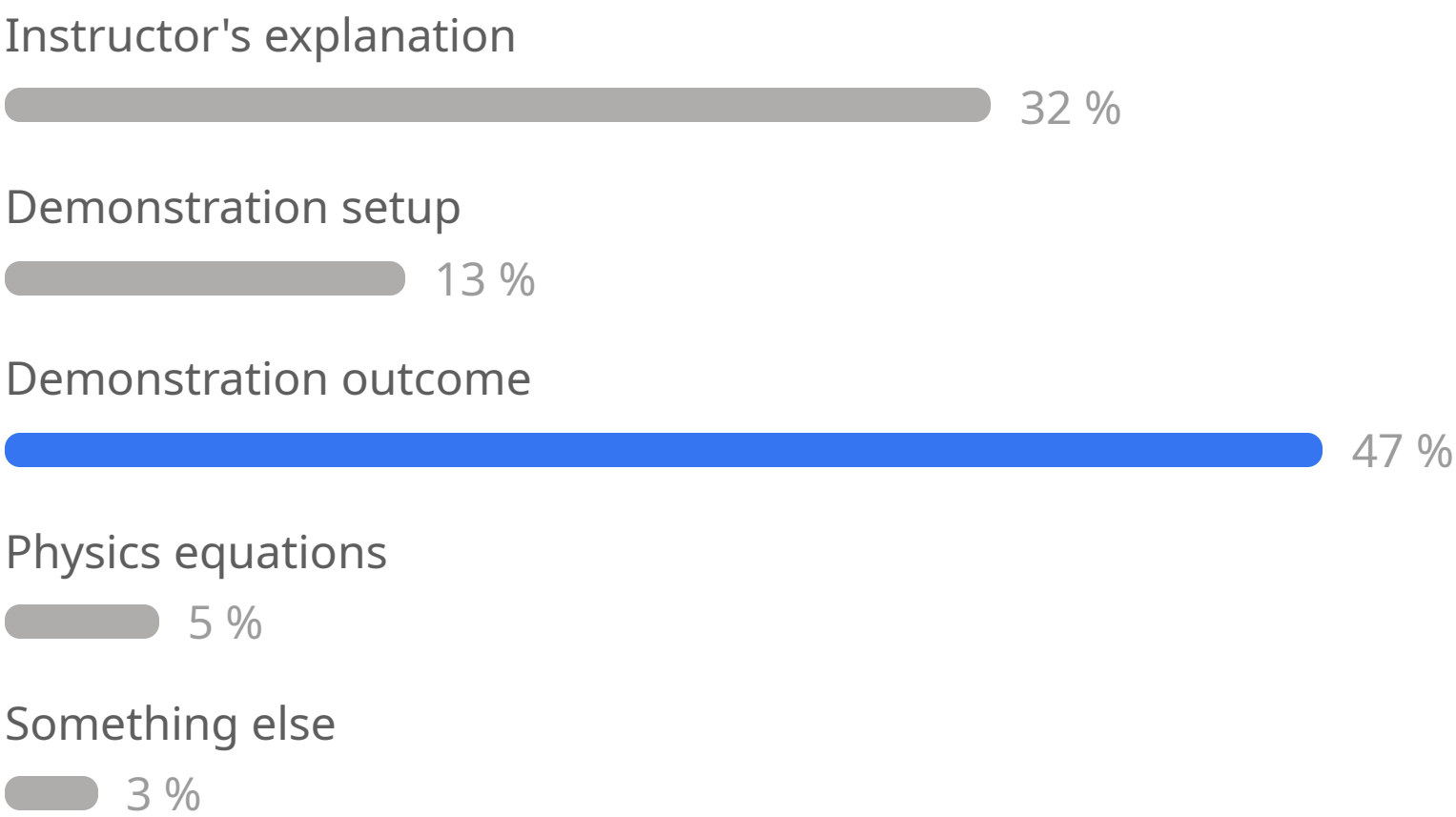


Multiple choice poll

9am Demo 2 POST: Laws of Motion (3/3)

0 3 8

During the demonstration, which aspect were you focused on the most?



9am Demo PRE: Circular Motion (1/2)

0 3 5

In this demo, an object is held at an angle from vertical. It is connected to the ceiling by a string. How should the instructor release the object such that it will travel in a circular path?

release the object from rest

0 %

push the object directly toward the center of the imaginary circle (toward vertical)

3 %

push the object directly away from the center of the imaginary circle (increasing the angle from vertical)

74 %

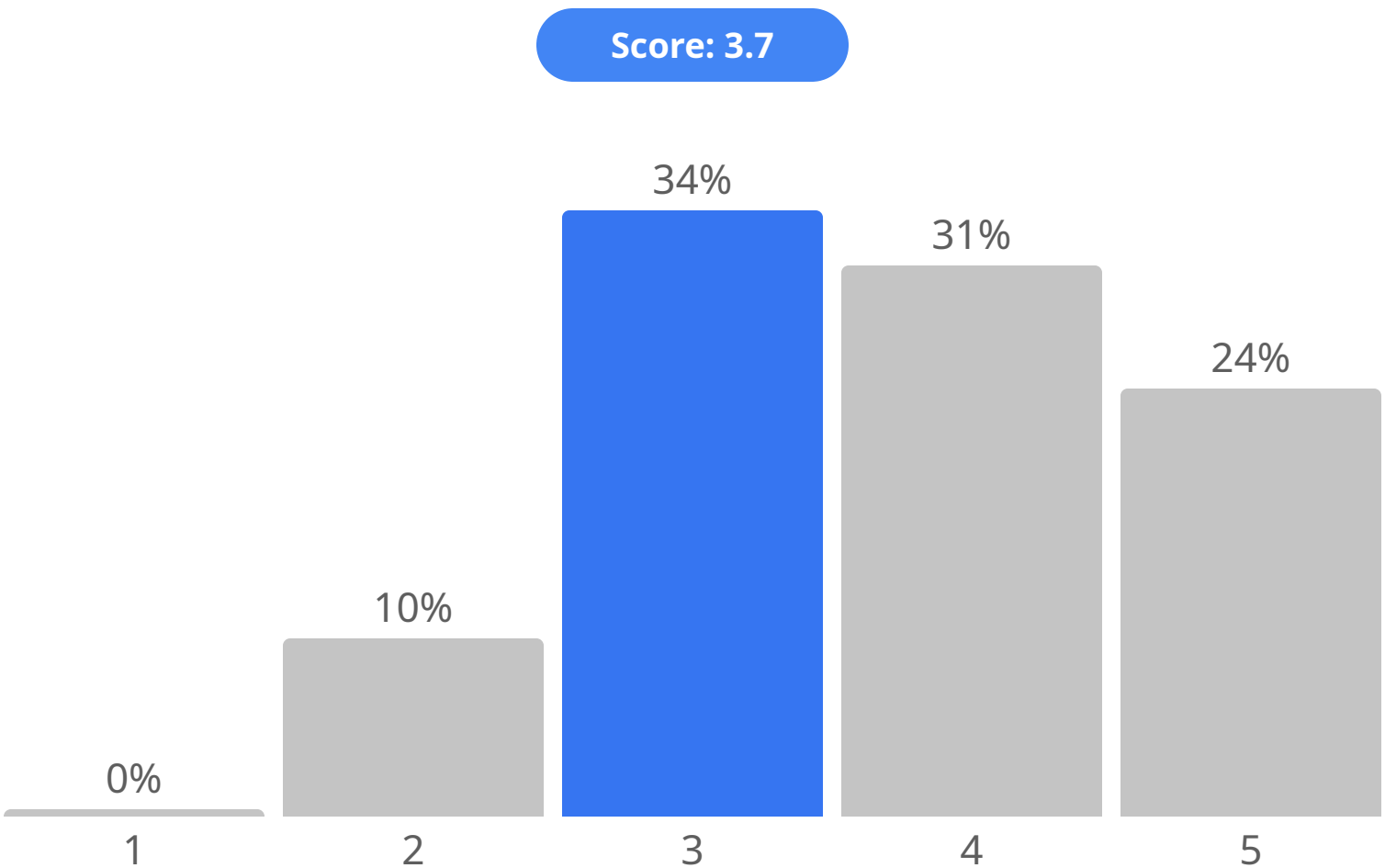
none of the above

23 %

9am Demo PRE: Circular Motion (2/2)

0 2 9

On a scale from 1-5, rate your confidence in your answer.



9am Demo 3 POST: Circular Motion (1/3)

0 3 4

In this demo, an object is held at an angle from vertical. It is connected to the ceiling by a string. When the object is released, which combination of initial velocity and acceleration will result in circular motion?

tangential initial velocity, tangential acceleration

9 %

tangential initial velocity, radial acceleration

76 %

radial initial velocity, tangential acceleration

12 %

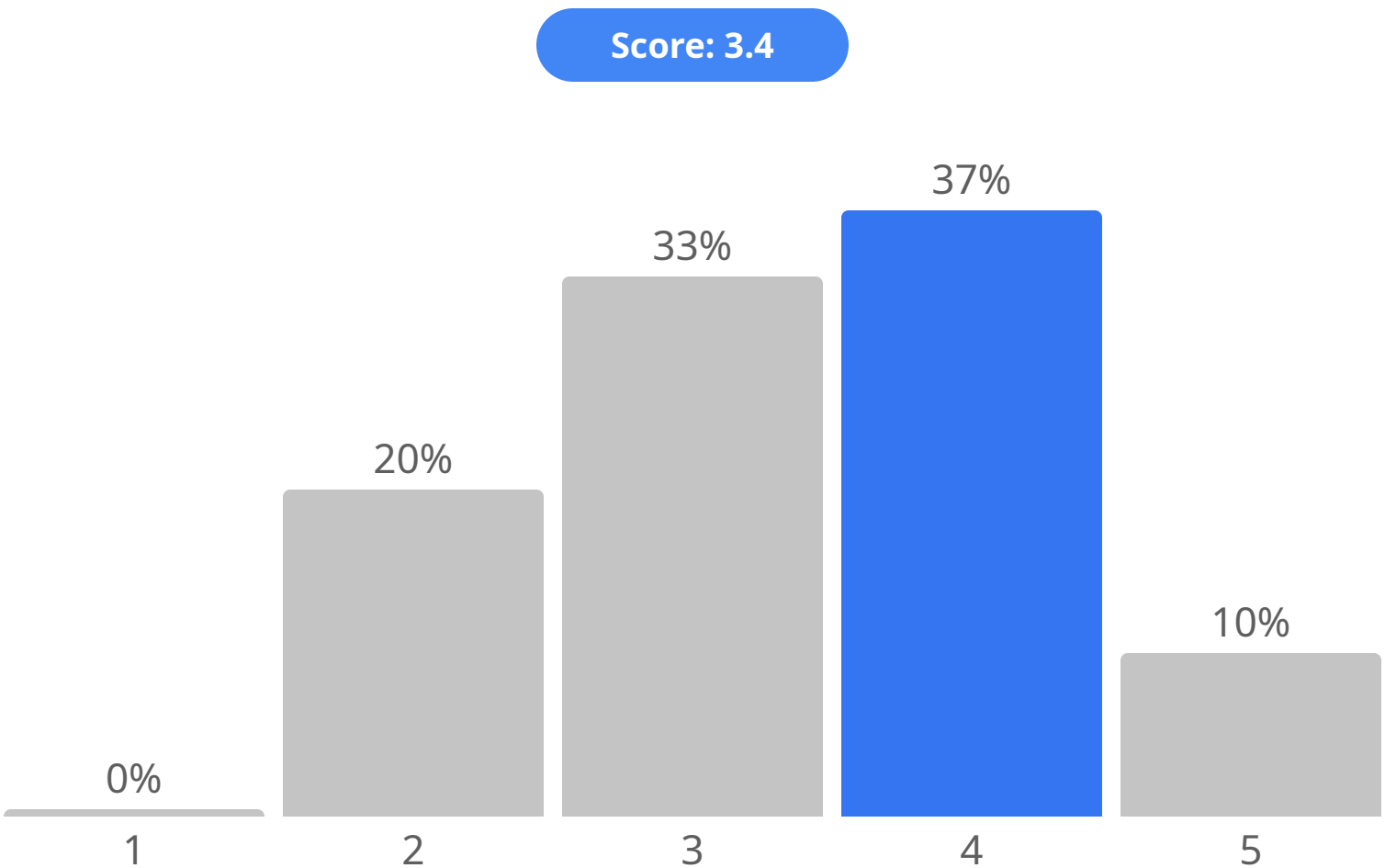
radial initial velocity, radial acceleration

3 %

9am Demo 3 POST: Circular Motion (2/3)

0 3 0

On a scale from 1-5, rate your confidence in your answer.

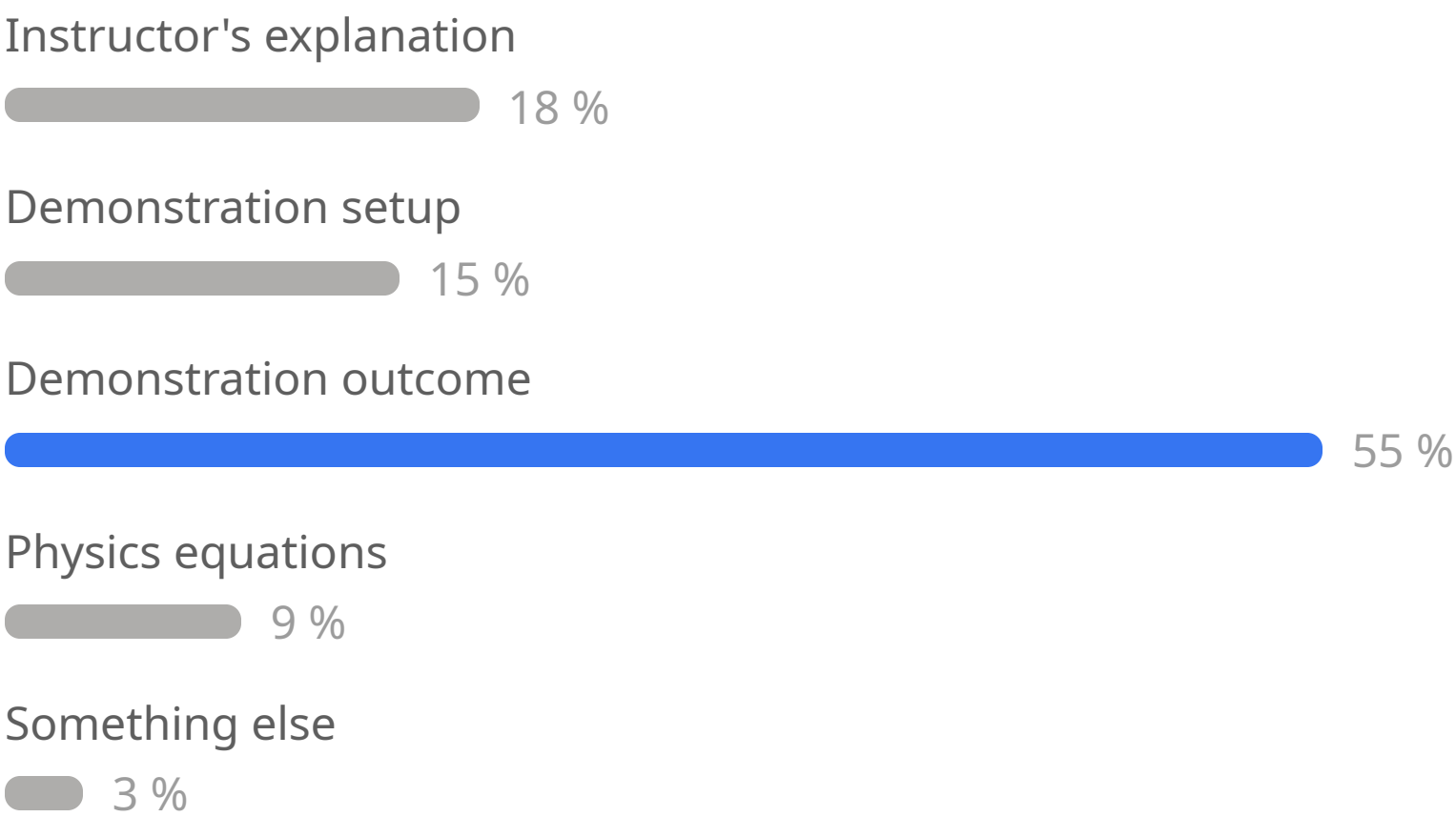


Multiple choice poll

9am Demo 3 POST: Circular Motion (3/3)

0 3 3

During the demonstration, which aspect were you focused on the most?



9am Demo 4 PRE: Loop-de-Loop (1/2)

0 3 0

In this demo, a ball will roll down a ramp into a vertical loop-de-loop track. If the ball is to complete the circular path, the ramp must be:

Shorter than the circular loop

0 %

As tall as the circular loop

0 %

Taller than the circular loop

93 %

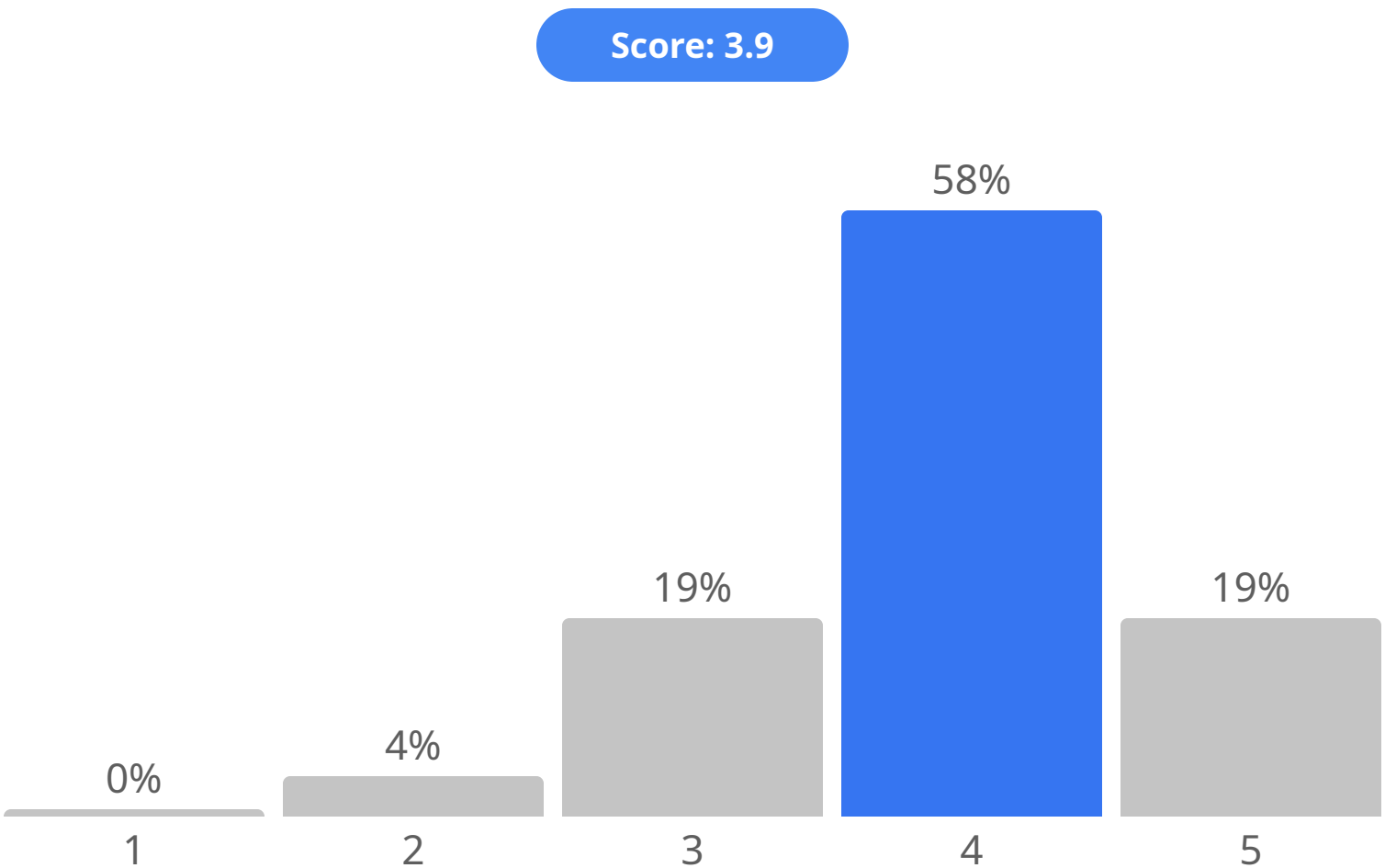
Two of these answers will work

7 %

9am Demo 4 PRE: Loop-de-Loop (2/2)

0 2 6

On a scale from 1-5, rate your confidence in your answer.



9am Demo 4 POST: Loop-de-Loop (1/3)

0 2 9

In this demo, a ball will roll down a ramp into a vertical loop-de-loop track. Modeling the motion of the ball completing the circular path, the required centripetal force will always:

Change the speed of the ball

0 %

Point toward the top of the loop

7 %

Be lesser in magnitude than the force of gravity

3 %

Point toward the center of the loop circle

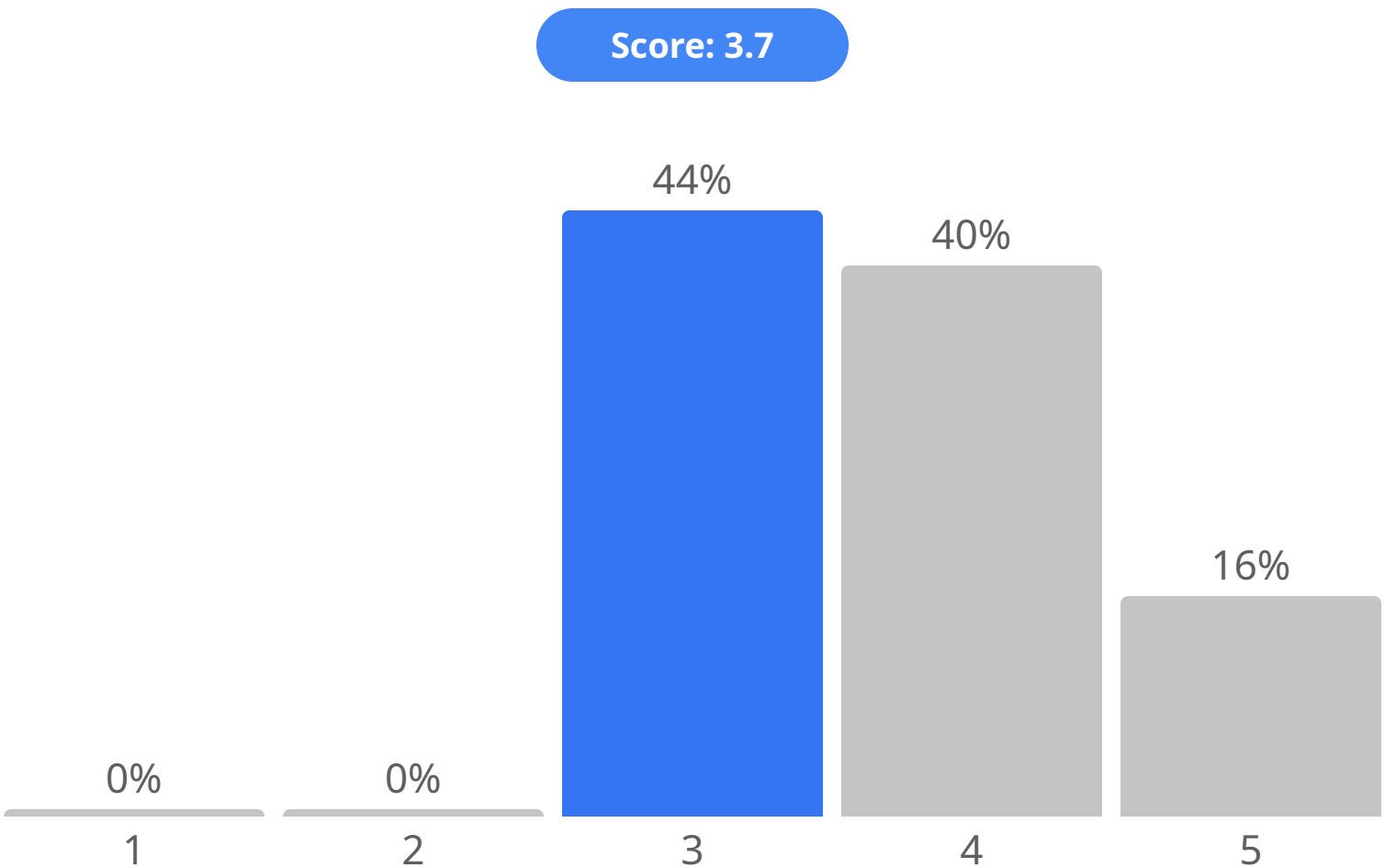
90 %

Rating poll

9am Demo 4 POST: Loop-de-Loop (2/3)

0 2 5

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

9am Demo 4 POST: Loop-de-Loop (3/3)

0 3 0

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

33 %

Demonstration setup

10 %

Demonstration outcome

50 %

Physics equations

3 %

Something else

3 %

9am Demo 5 PRE: Pendulum (1/2)

0 2 9

In this demo, a bowling ball is suspended from the ceiling via a rope, just like a pendulum. The instructor will hold the ball up to his chest and then release it. Which of the following statements about the ball's mechanical energy is true:
(1/2)

The ball's potential energy is greatest at the lowest point of its trajectory.

7 %

If the energy is conserved, potential and kinetic energy will change by equal and opposite values.

83 %

The ball's kinetic energy is greatest at the highest point of its trajectory.

7 %

9am Demo 5 PRE: Pendulum (1/2)

0 2 9

In this demo, a bowling ball is suspended from the ceiling via a rope, just like a pendulum. The instructor will hold the ball up to his chest and then release it. Which of the following statements about the ball's mechanical energy is true:
(2/2)

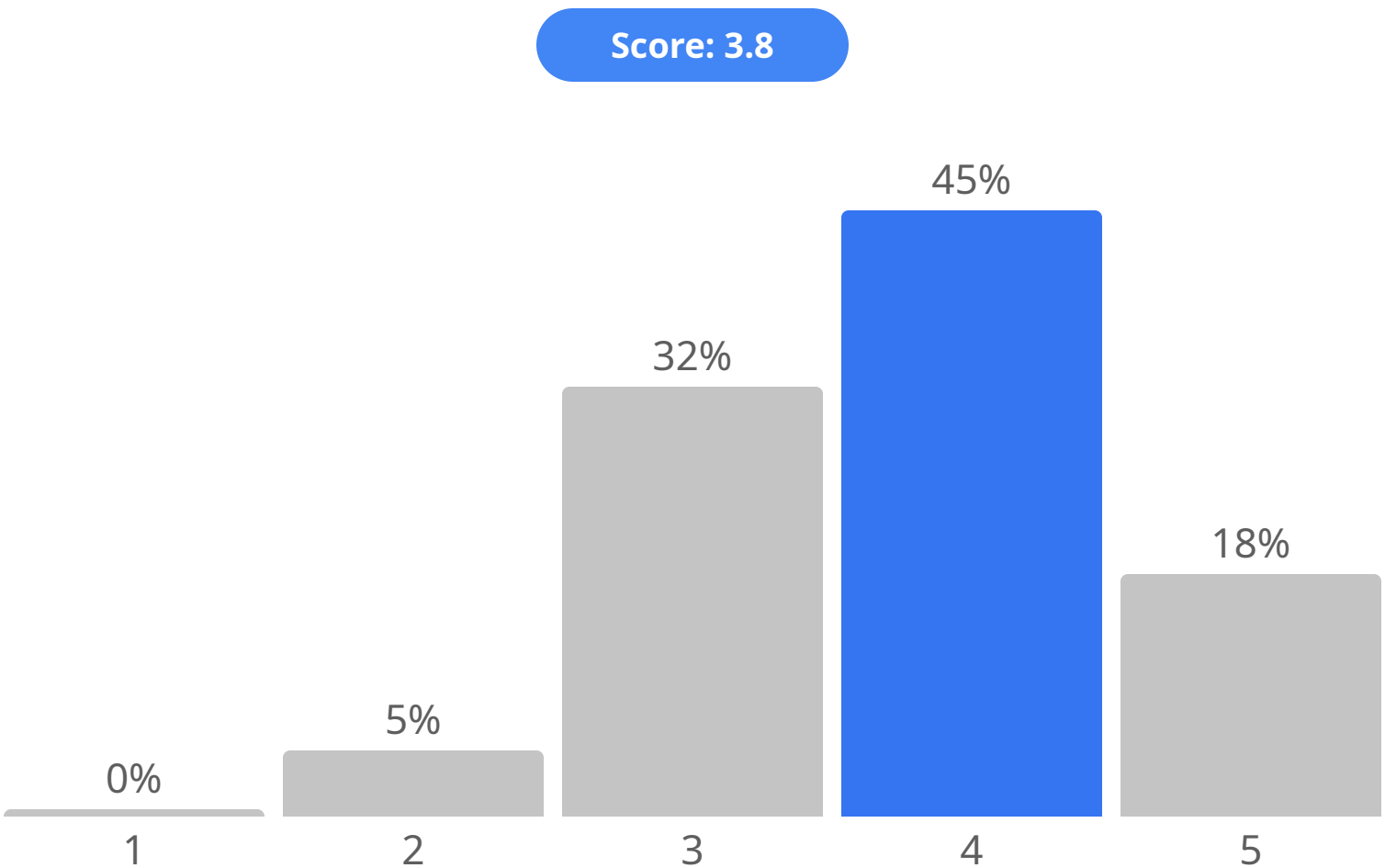
If the energy is conserved, the potential energy will be constant.

3 %

9am Demo 5 PRE: Pendulum (2/2)

0 2 2

On a scale from 1-5, rate your confidence in your answer.



9am Demo 5 POST: Pendulum (1/3)

0 2 8

In this demo, a bowling ball is suspended from the ceiling via a rope, just like a pendulum. The instructor will hold the ball up to his chest and then release it. The instructor can safely:

Release the ball from rest

96 %

Push the ball forward

4 %

Both of these

0 %

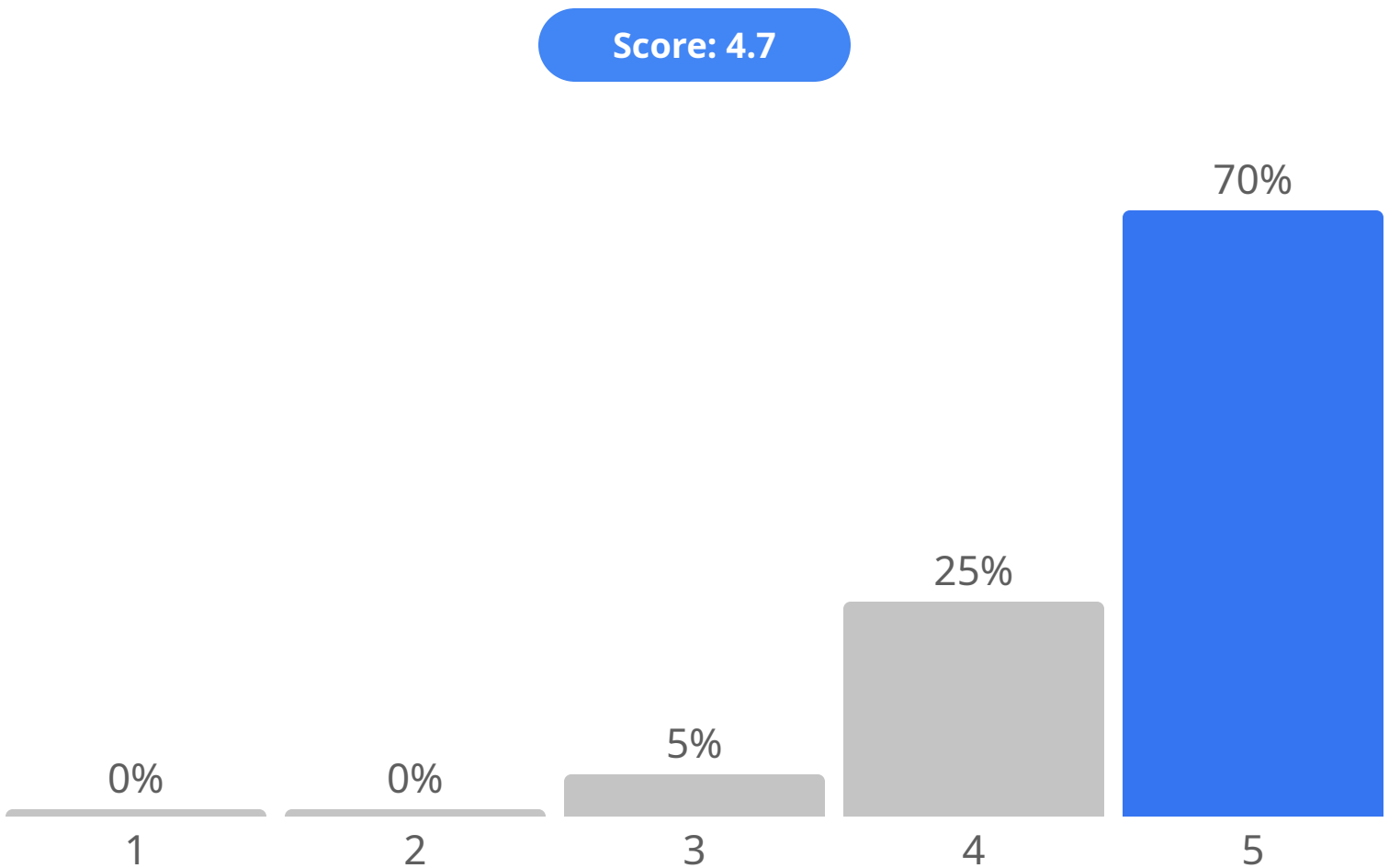
Neither of these

0 %

9am Demo 5 POST: Pendulum (2/3)

0 2 0

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

9am Demo 5 POST: Pendulum (3/3)

0 2 8

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

18 %

Demonstration setup

7 %

Demonstration outcome

64 %

Physics equations

4 %

Something else

7 %

Multiple choice poll

9am Demo 6 PRE: Collisions (1/2)

0 3 8

This demo involves elastic and inelastic collisions. For inelastic collisions, what is conserved?

Total Kinetic Energy of the System

18 %

Total Momentum of the System

61 %

Both

13 %

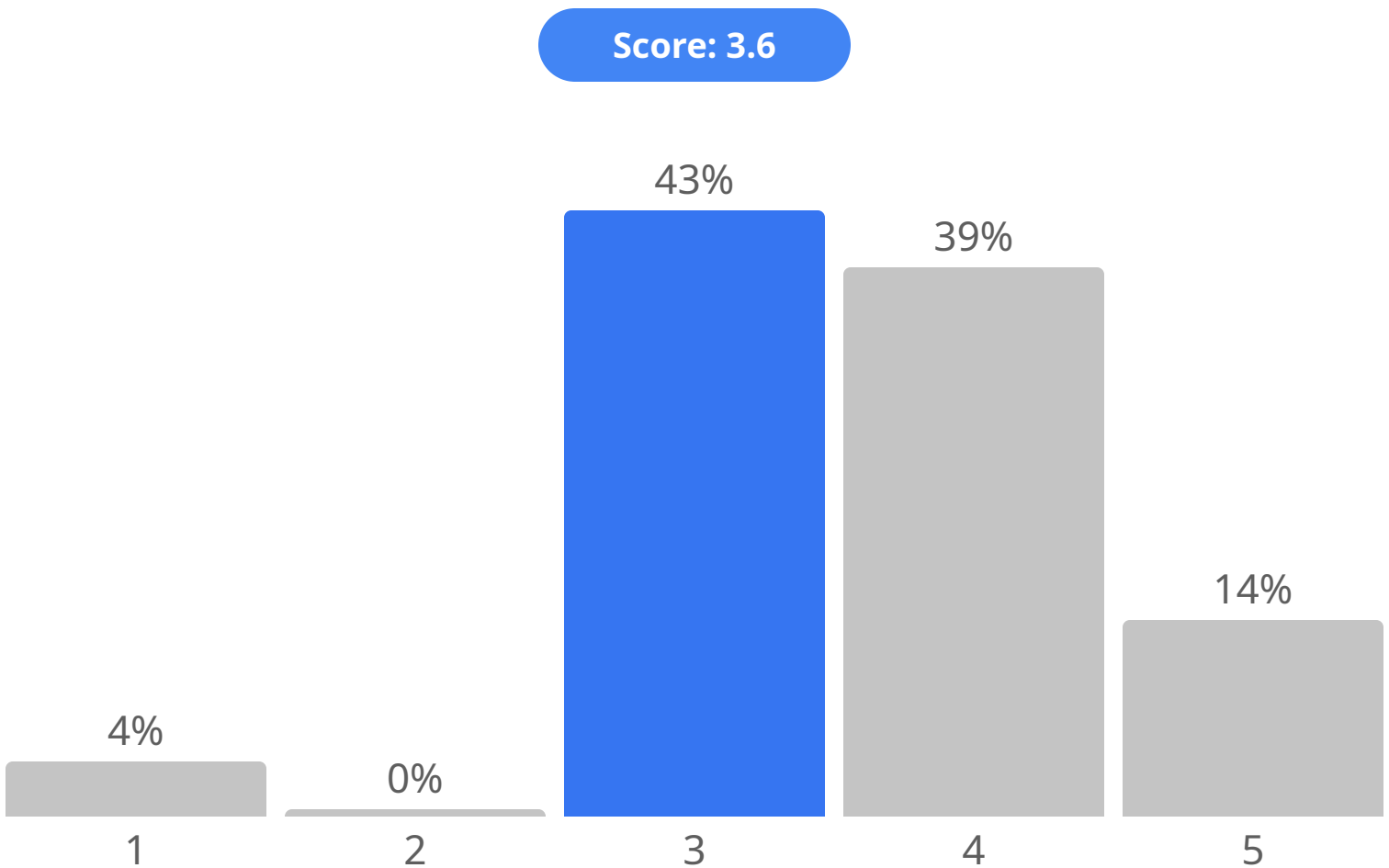
Neither

8 %

9am Demo 6 PRE: Collisions (2/2)

0 2 8

On a scale from 1-5, rate your confidence in your answer.



9am Demo 6 POST: Collisions (1/3)

0 3 8

Two balls of equal mass and different elasticity will roll down a ramp (from the same height). At the bottom of the ramp is an upright block of similar mass. Upon collision, which ball has a higher chance of toppling the block?

Happy (more elastic) ball

82 %

Sad (more inelastic) ball

11 %

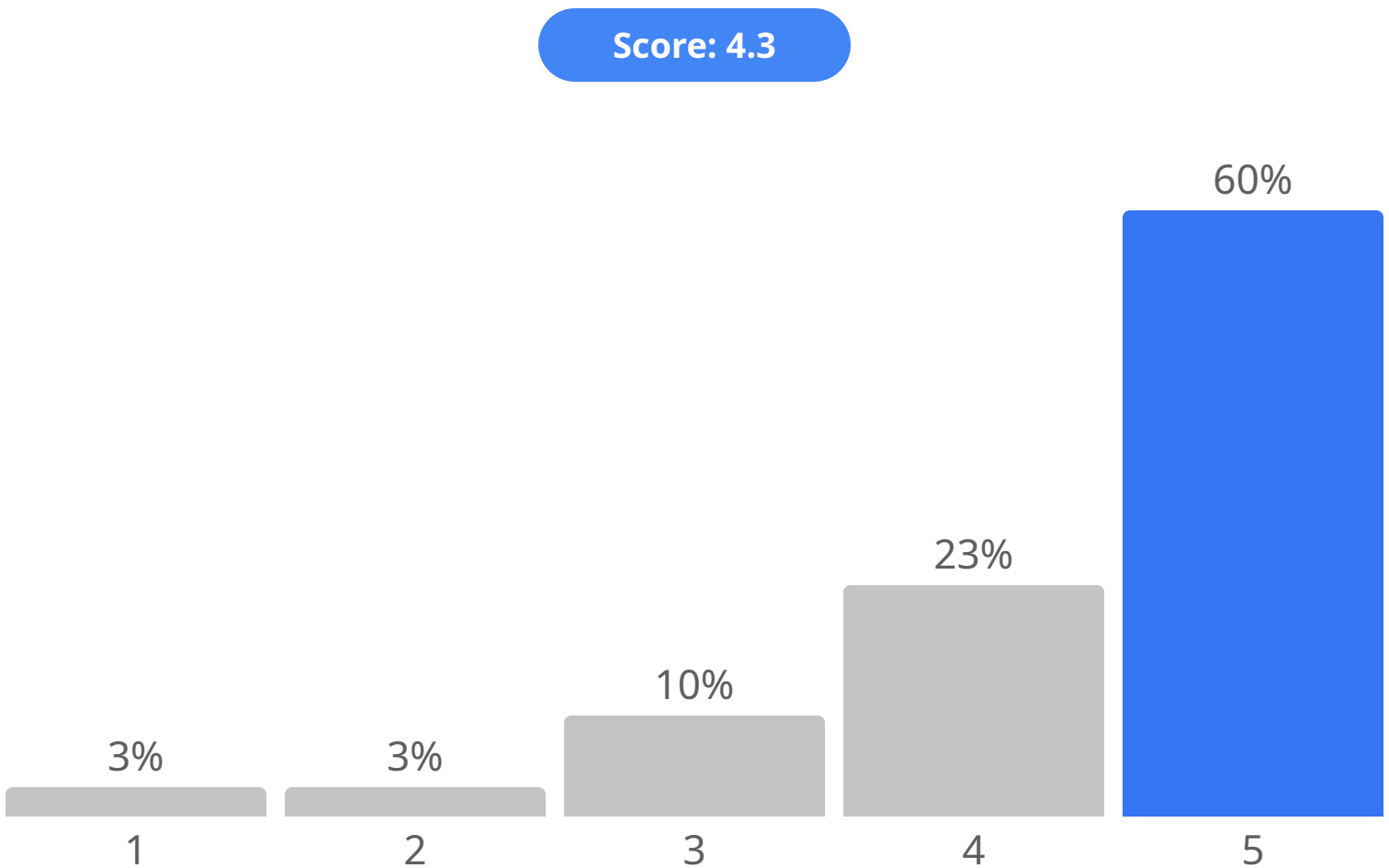
They have an equal chance

8 %

9am Demo 6 POST: Collisions (2/3)

0 3 0

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

9am Demo 6 POST: Collisions (3/3)

0 3 8

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

8 %

Demonstration setup

3 %

Demonstration outcome

82 %

Physics equations

3 %

Something else

5 %

9am Demo 7 PRE: Rolling Down a Ramp (1/2)

0 3 6

In this demo, a sphere will roll down a ramp. Which of the following quantities will affect the time it takes the sphere to reach the bottom?

Mass of the sphere

0 %

Radius of the sphere

11 %

Both

86 %

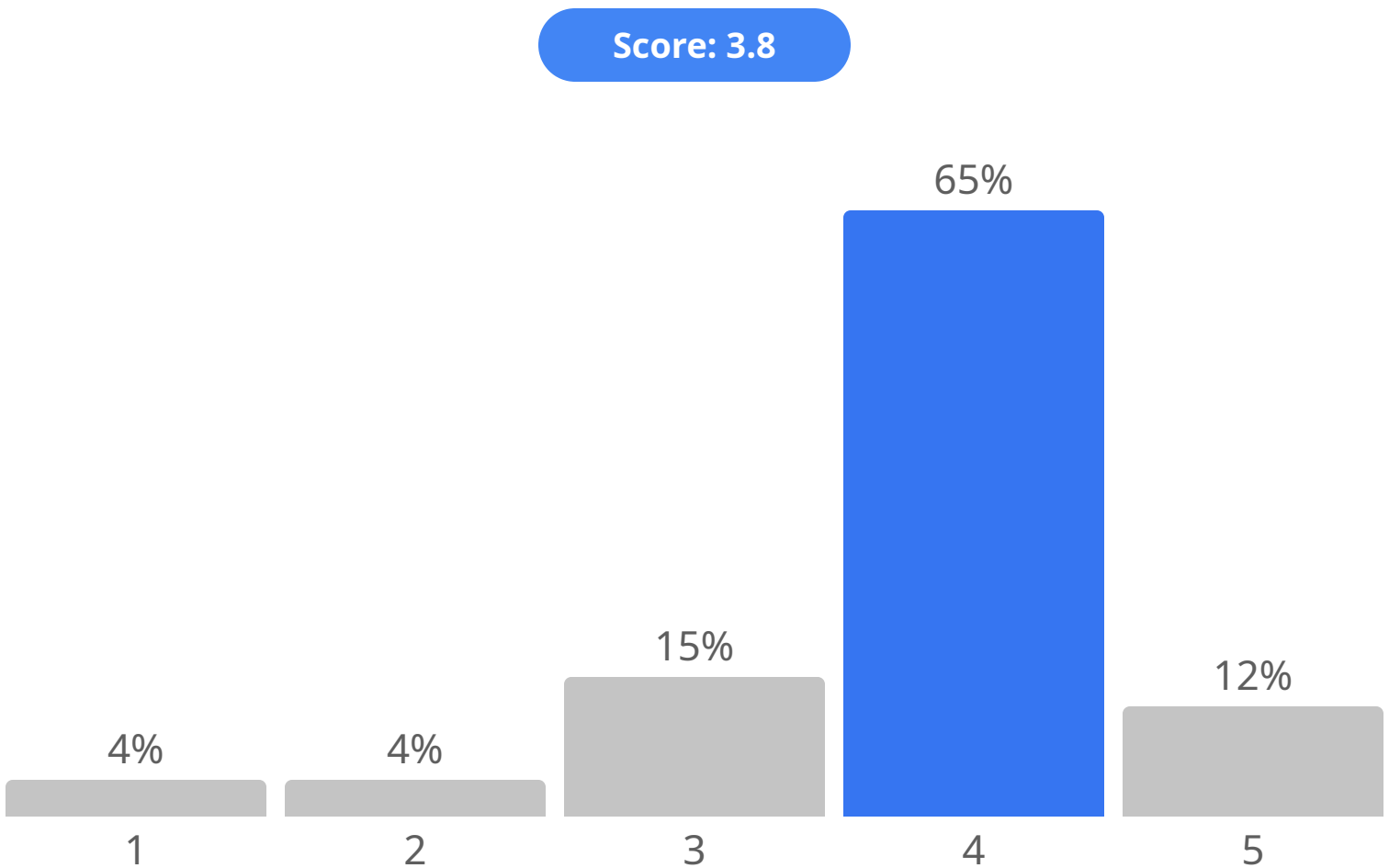
Neither

3 %

9am Demo 7 PRE: Rolling Down a Ramp (2/2)

0 2 6

On a scale from 1-5, rate your confidence in your answer.



9am Demo 7 POST: Rolling Down a Ramp (1/3)

0 3 5

In this demo, a solid sphere and a solid cylinder will roll down a ramp (released from rest at the same height). Which one will reach the bottom first?

They will reach the bottom at the same time

80 %

The sphere will reach the bottom first

9 %

The cylinder will reach the bottom first

9 %

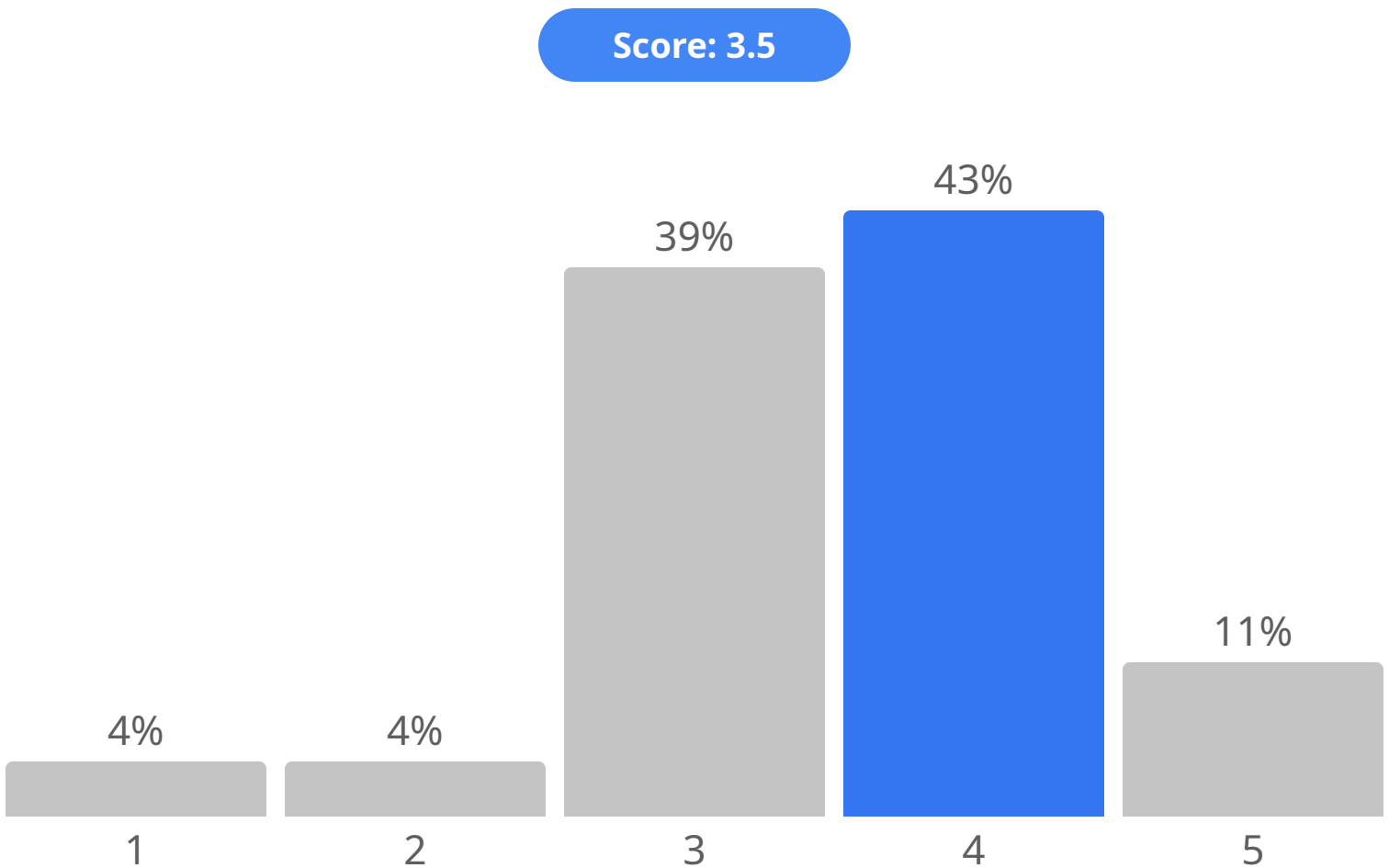
It will depend on the mass of each object

3 %

9am Demo 7 POST: Rolling Down a Ramp (2/3)

0 2 8

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

9am Demo 7 POST: Rolling Down a Ramp (3/3)

0 3 4

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

12 %

Demonstration setup

9 %

Demonstration outcome

68 %

Physics equations

12 %

Something else

0 %

9am Demo 8 PRE: Dumbbell Torque (1/2)

0 2 9

In this demo, a participant sits on a swiveling chair while holding a dumbbell in each hand. They begin with their arms out wide and then bring the dumbbells into their chest. Which of the following is NOT true:

The participant exerts a torque

62 %

The participant causes an angular acceleration on the system

10 %

The participant preserves the kinetic energy of the system

7 %

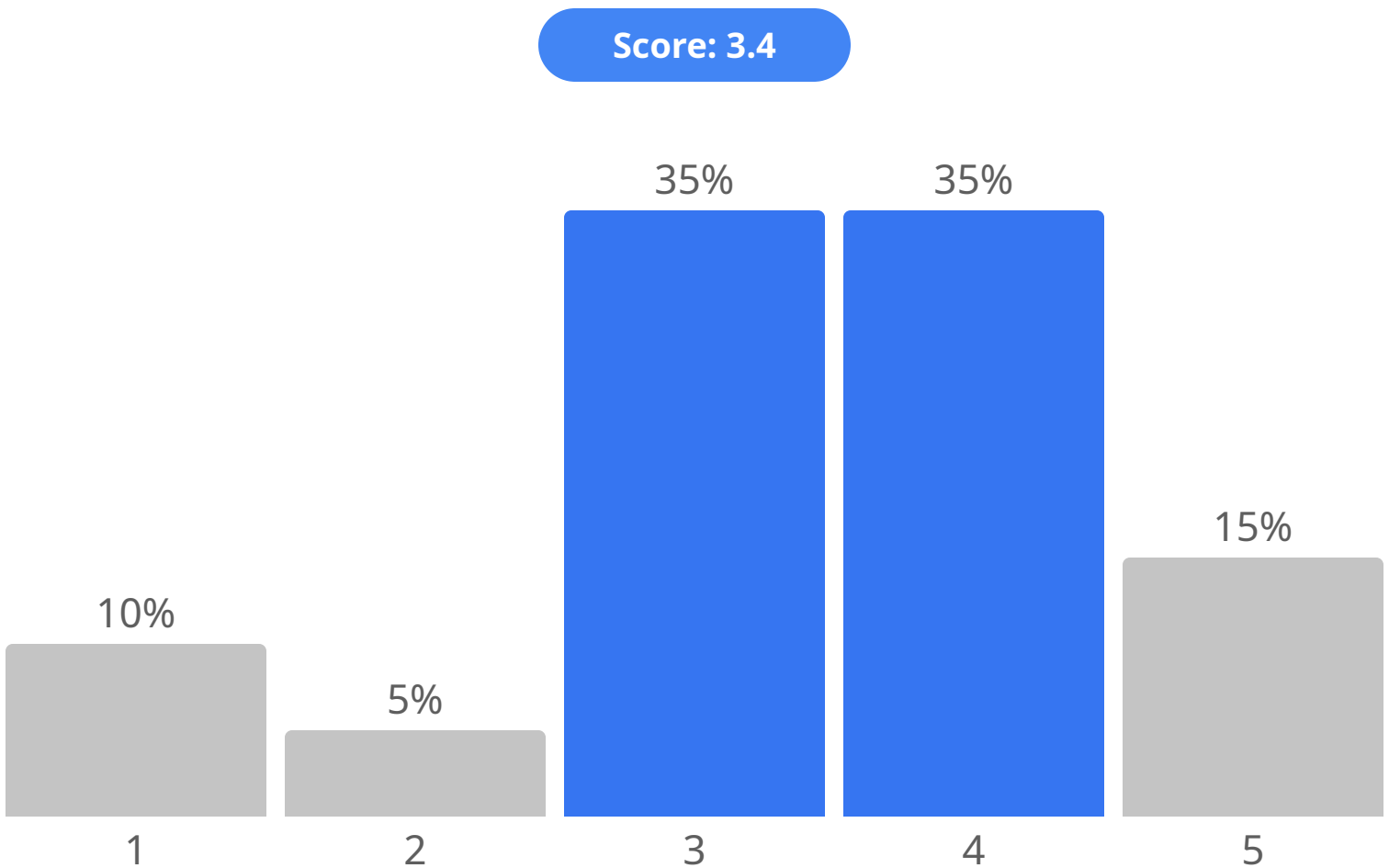
The participant changes the rotational inertia of the system

21 %

9am Demo 8 PRE: Dumbbell Torque (2/2)

0 2 0

On a scale from 1-5, rate your confidence in your answer.



9am Demo 8 POST: Dumbbell Torque (1/3)

0 2 6

In this demo, a participant sits on a swiveling chair while holding a dumbbell in each hand. They begin with their arms out wide and then bring the dumbbells into their chest. Upon doing this, their angular velocity will:

Increase

100 %

Decrease

0 %

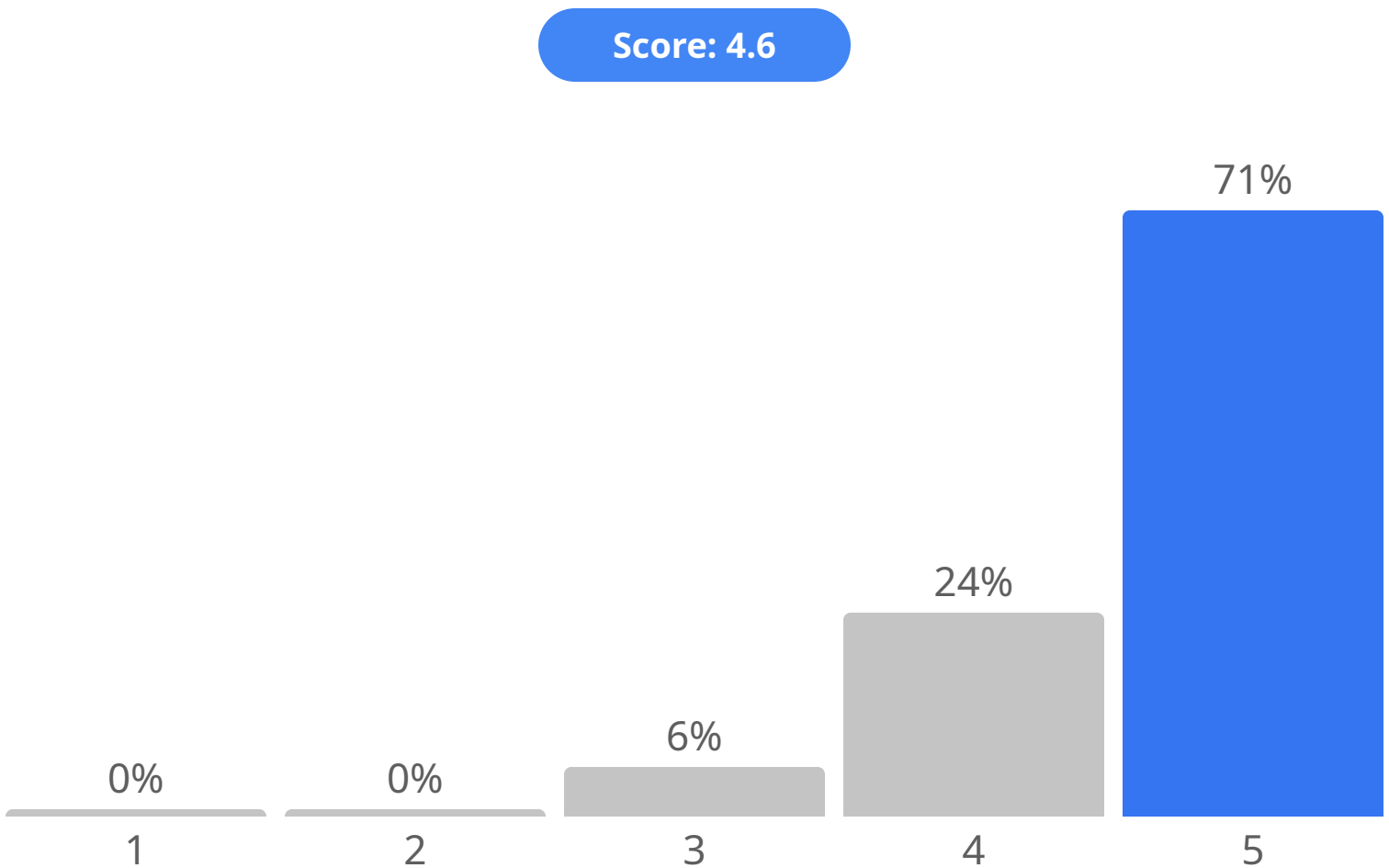
Stay the same

0 %

9am Demo 8 POST: Dumbbell Torque (2/3)

0 1 7

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

9am Demo 8 POST: Dumbbell Torque (3/3)

0 2 6

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

12 %

Demonstration setup

8 %

Demonstration outcome

65 %

Physics equations

8 %

Something else

8 %

PM Surveys

Poll results

slido

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- 2pm Demo 5 PRE: Pendulum
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2pm Demo 1 PRE: Projectile Motion (1/2)

0 3 9

In this demo, three students are arranged in a straight line. One student throws a basketball in an arc to the first (closer) student, and then to the second (farther) student.
(1/2)

The kinematic equations cannot be used to analyze this problem, because acceleration will be constant.

3 %

The kinematic equations cannot be used to analyze this problem, because acceleration will not be constant.

21 %

The kinematic equations can be used to analyze this problem, because acceleration will be constant.

67 %

2pm Demo 1 PRE: Projectile Motion (1/2)

0 3 9

In this demo, three students are arranged in a straight line. One student throws a basketball in an arc to the first (closer) student, and then to the second (farther) student.
(2/2)

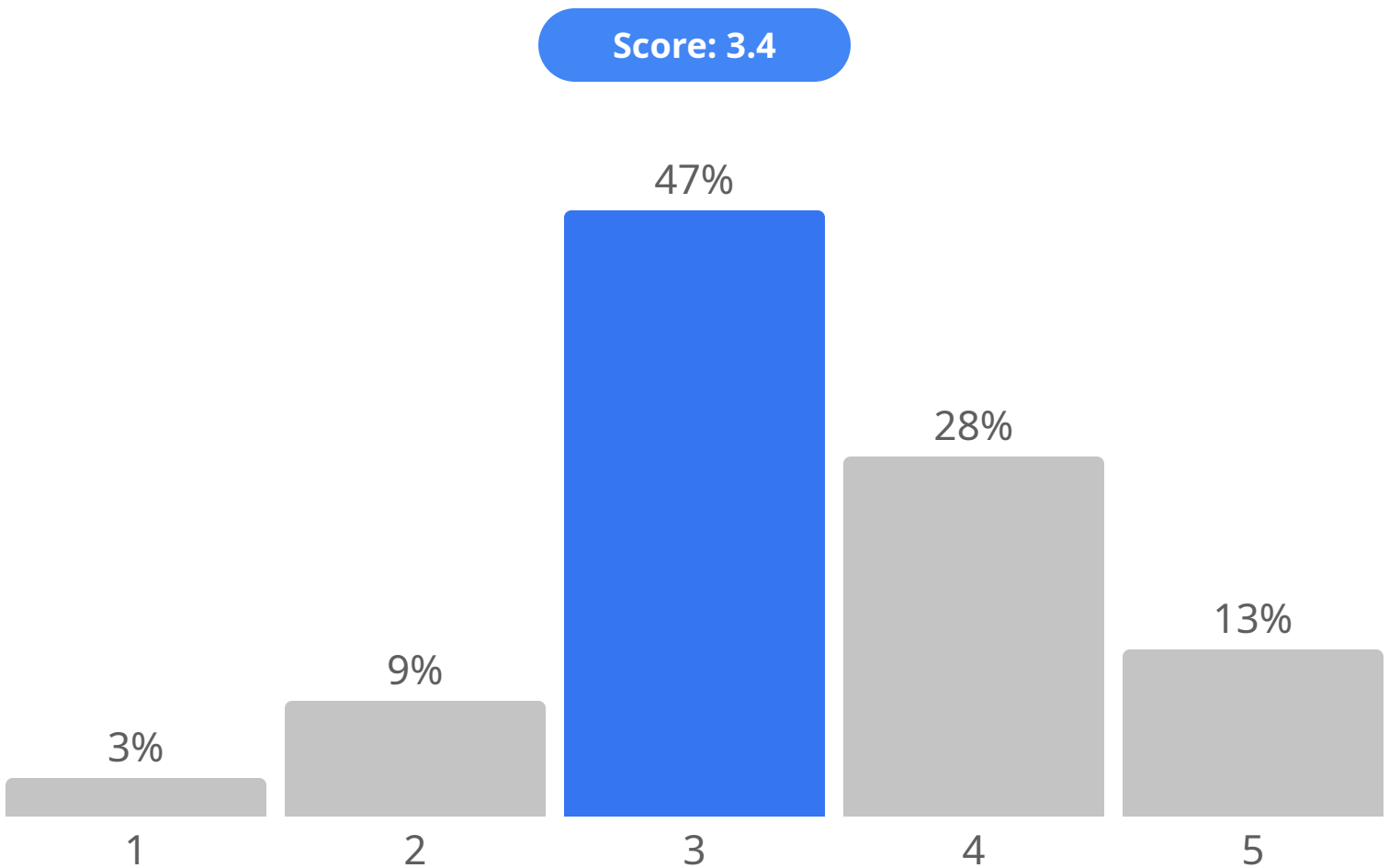
The kinematic equations can be used to analyze this problem, because acceleration will not be constant.

10 %

2pm Demo 1 PRE: Projectile Motion (2/2)

0 3 2

On a scale from 1-5, rate your confidence in your answer.



2pm Demo 1 POST: Projectile Motion (1/3)

0 3 9

In this demo, three students are arranged in a straight line. One student throws a basketball in an arc to the first (closer) student, and then to the second (farther) student.
(1/2)

The student could adjust either the initial velocity or the angle of the basketball to complete the second pass.

85 %

The student must only adjust the angle of the basketball to complete the second pass.

5 %

The student must only adjust the initial velocity of the basketball to complete the second pass.

10 %

2pm Demo 1 POST: Projectile Motion (1/3)

0 3 9

In this demo, three students are arranged in a straight line. One student throws a basketball in an arc to the first (closer) student, and then to the second (farther) student.

(2/2)

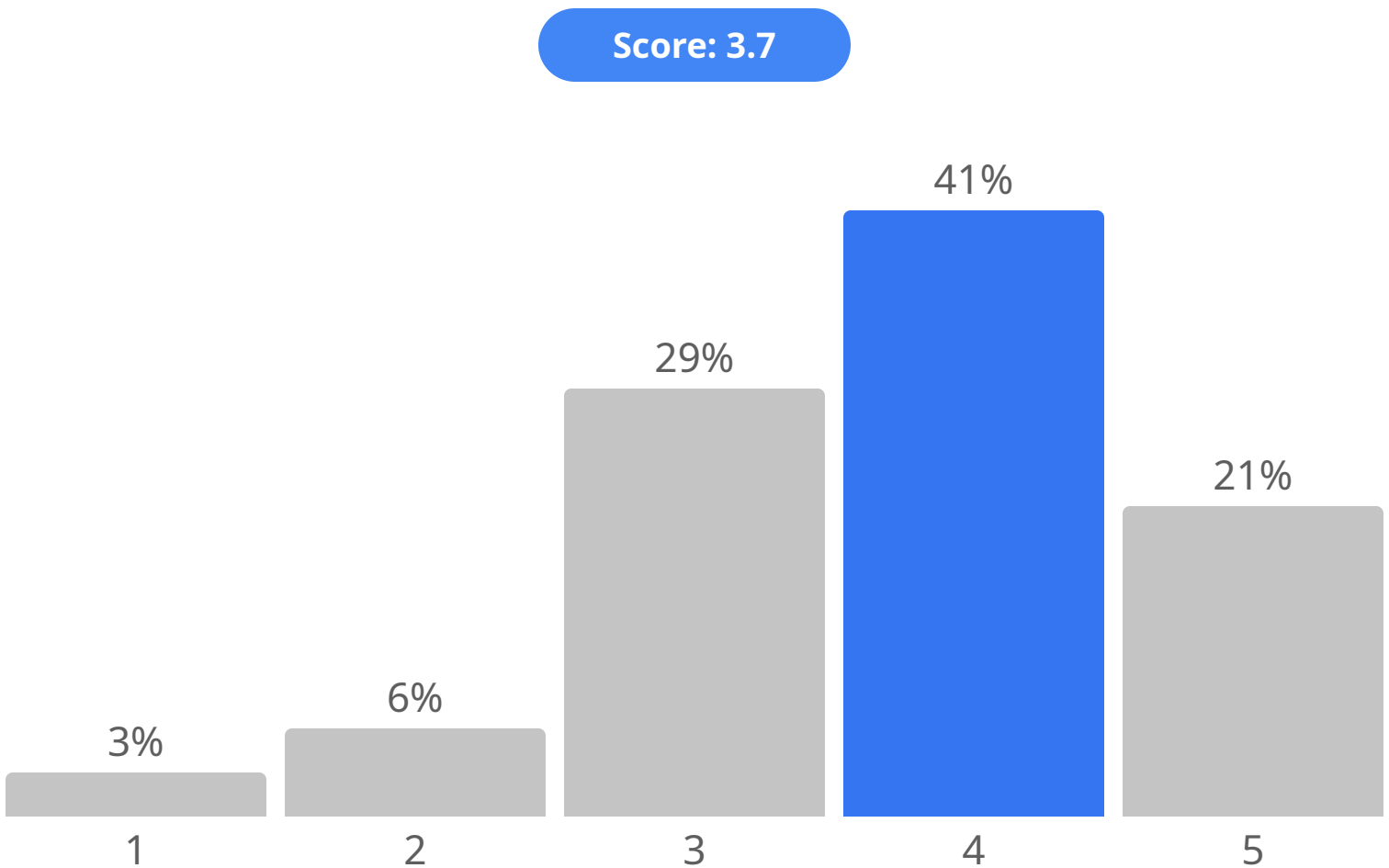
The student needs to adjust something else to complete the second pass.

0 %

2pm Demo 1 POST: Projectile Motion (2/3)

0 3 4

On a scale from 1-5, rate your confidence in your answer.

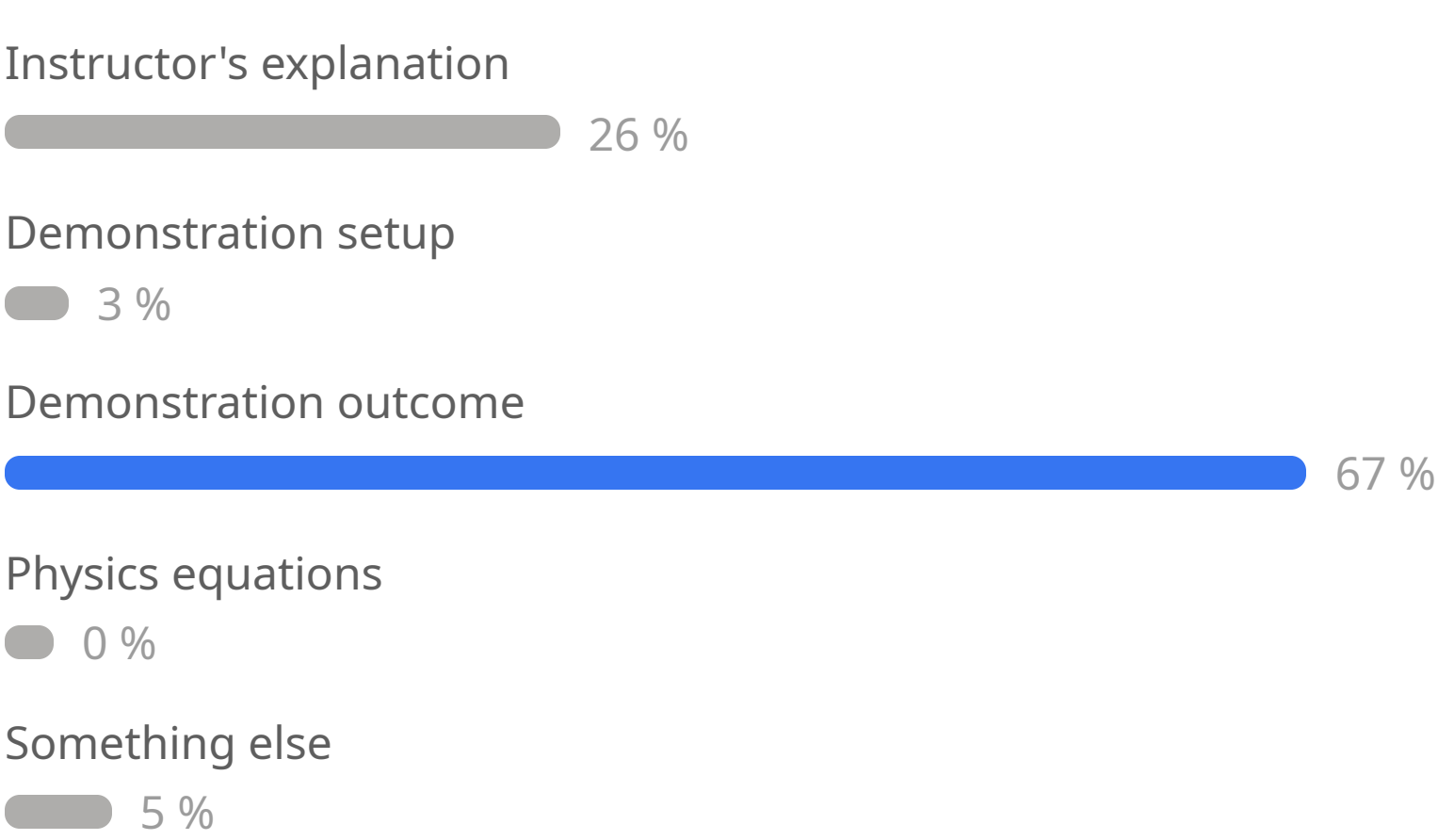


Multiple choice poll

2pm Demo 1 POST: Projectile Motion (3/3)

0 3 9

During the demonstration, which aspect were you focused on the most?



2pm Demo 2 PRE: Laws of Motion (1/2)

0 4 0

In this demo, two objects of comparable mass are attached by a string, which runs through a pulley. One object sits atop a table, while the other is suspended in the air. The suspended object will be released from rest.
(1/2)

The hanging mass will not accelerate at all, it will fall with constant speed.

13 %

The hanging mass will fall with acceleration smaller than 9.8 m/s^2, due to the object on the table.

83 %

The hanging mass will fall with acceleration 9.8 m/s^2, because the mass on the table does not affect acceleration.

3 %

2pm Demo 2 PRE: Laws of Motion (1/2)

0 4 0

In this demo, two objects of comparable mass are attached by a string, which runs through a pulley. One object sits atop a table, while the other is suspended in the air. The suspended object will be released from rest.
(2/2)

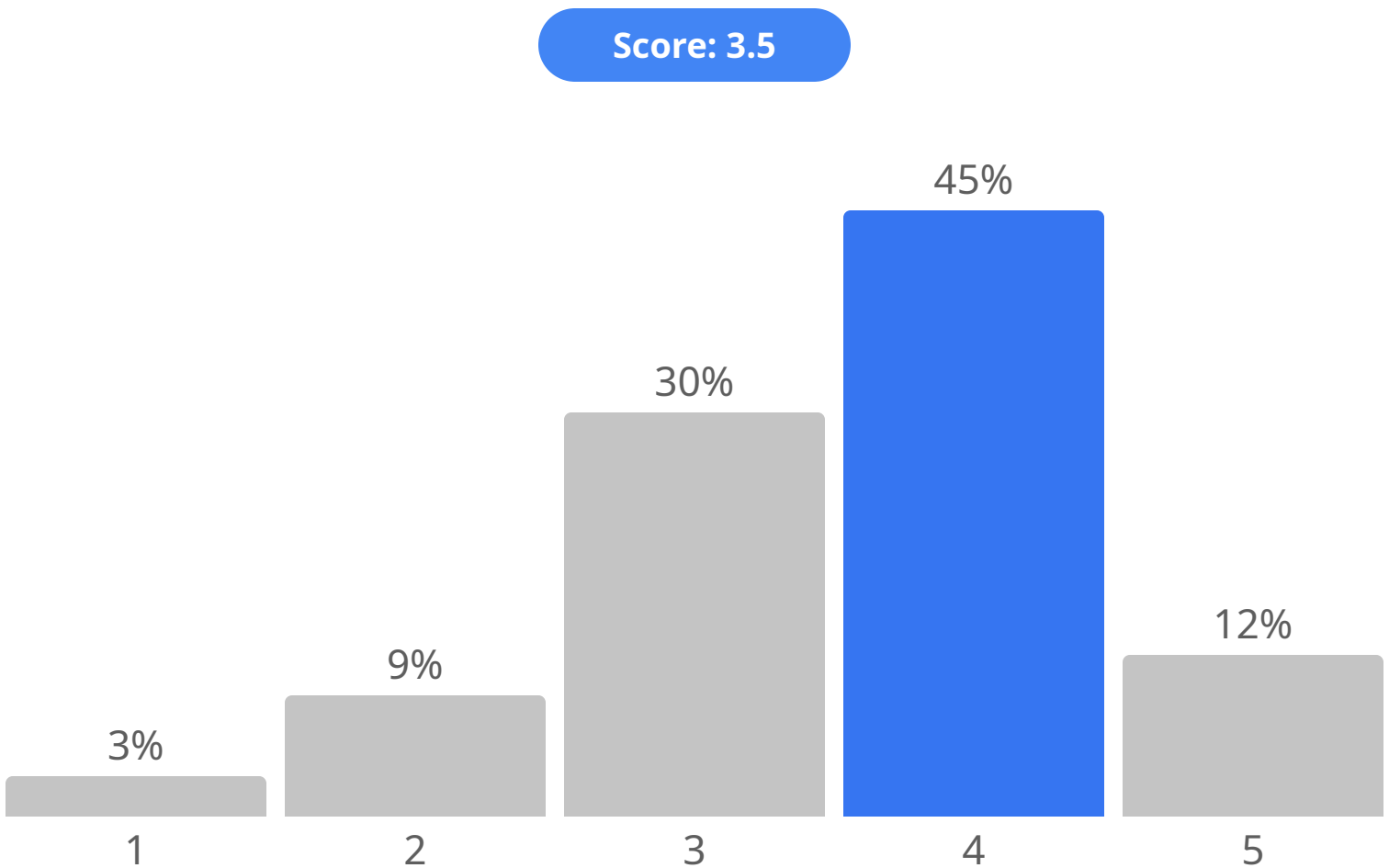
The hanging mass will fall with acceleration greater than 9.8 m/s^2, due to the object on the table.

3 %

2pm Demo 2 PRE: Laws of Motion (2/2)

0 3 3

On a scale from 1-5, rate your confidence in your answer.



2pm Demo 2 POST: Laws of Motion (1/3)

0 3 9

In this demo, two objects of comparable mass are attached by a string, which runs through a pulley. One object sits atop a table, while the other is suspended in the air. The suspended object will be released from rest.

Only the hanging mass will move.

3 %

The hanging mass starts falling, and after a few seconds the other mass moves.

5 %

The two masses both move, but the hanging mass will move faster.

8 %

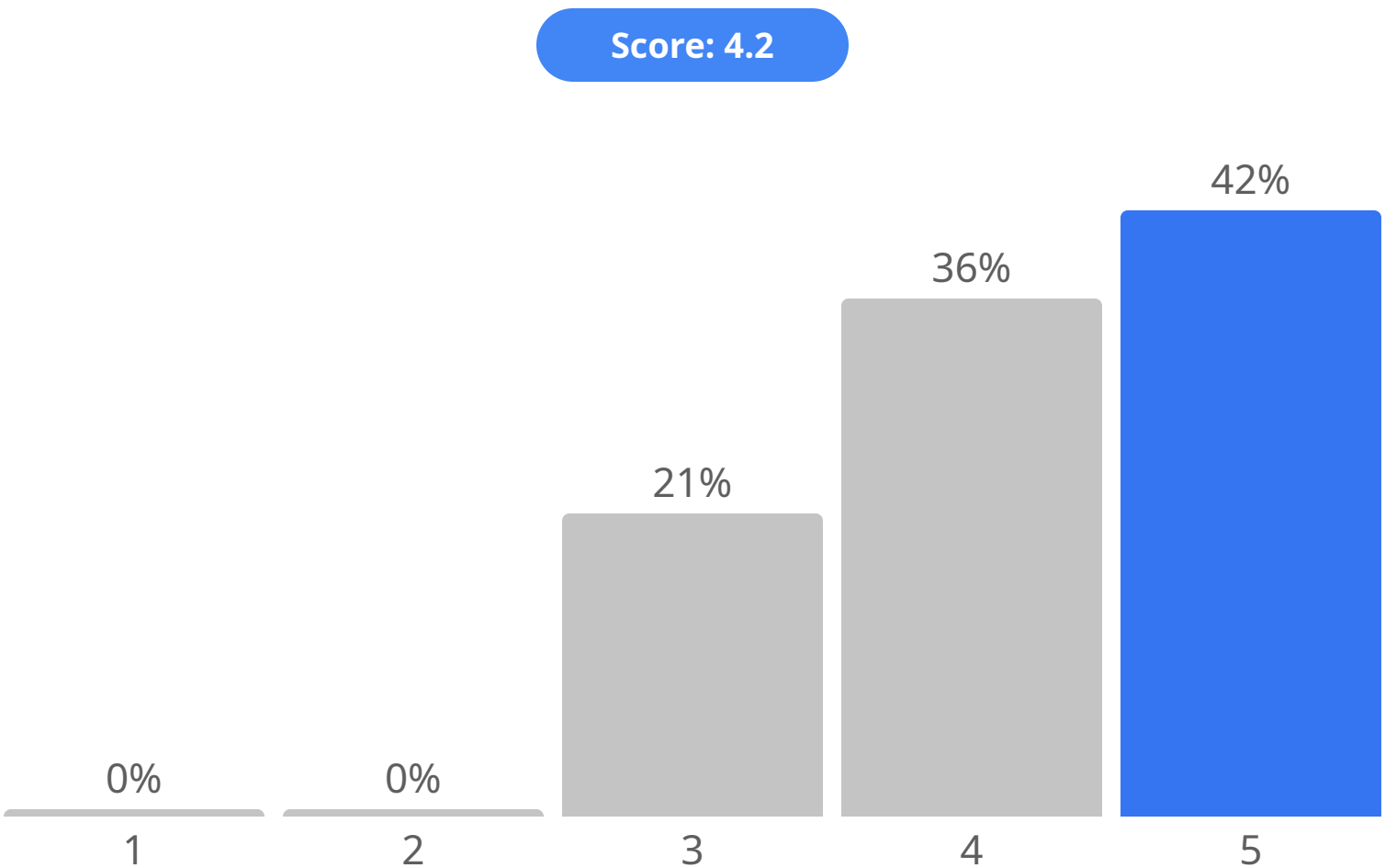
The two masses will move with the same speed.

85 %

2pm Demo 2 POST: Laws of Motion (2/3)

0 3 3

On a scale from 1-5, rate your confidence in your answer.

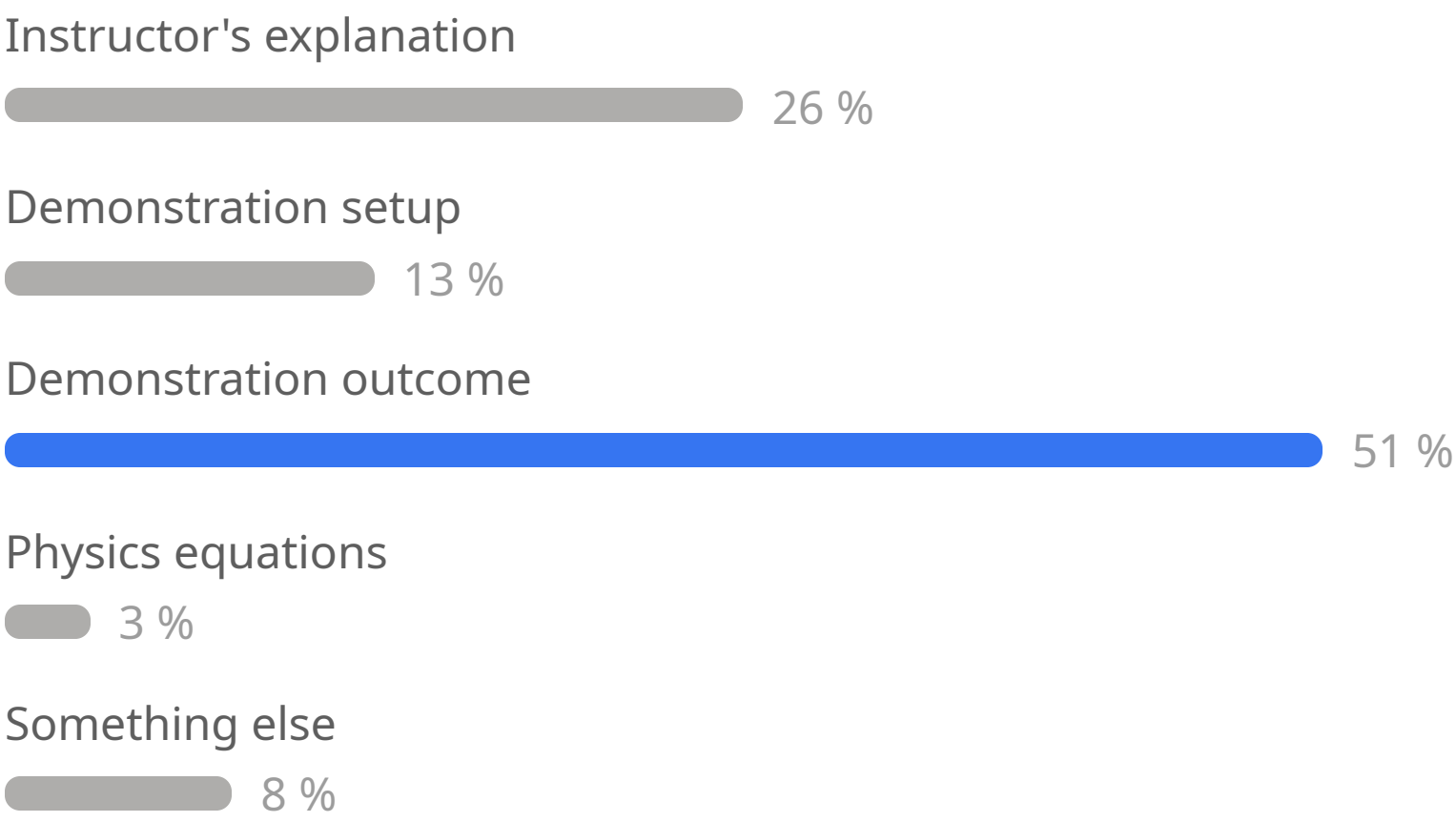


Multiple choice poll

2pm Demo 2 POST: Laws of Motion (3/3)

0 3 9

During the demonstration, which aspect were you focused on the most?



2pm Demo 3 PRE: Circular Motion (1/2)

0 3 8

In this demo, an object is held at an angle from vertical. It is connected to the ceiling by a string. When the object is released, which combination of initial velocity and acceleration will result in circular motion?

tangential initial velocity, tangential acceleration

0 %

tangential initial velocity, radial acceleration

84 %

radial initial velocity, tangential acceleration

16 %

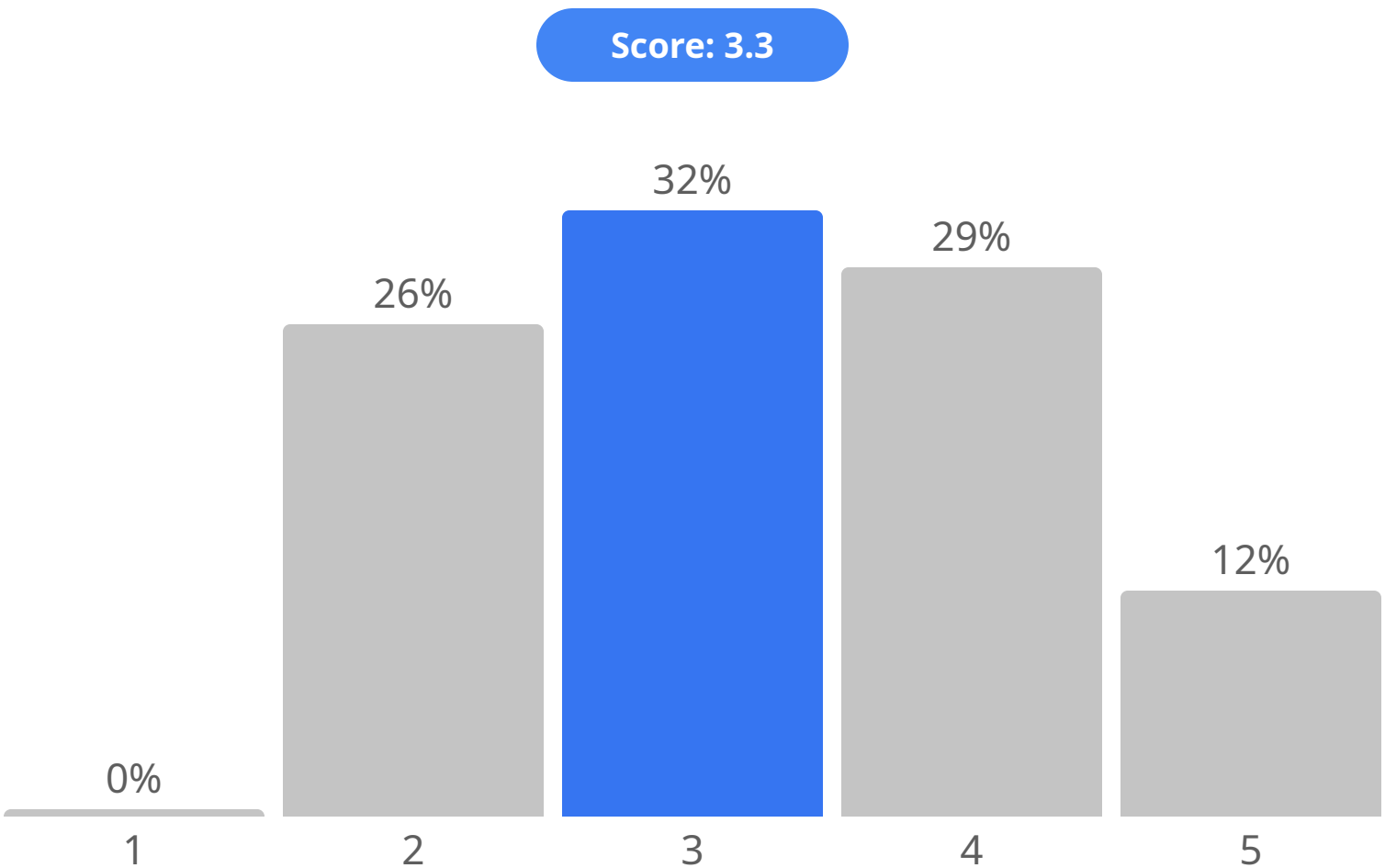
radial initial velocity, radial acceleration

0 %

2pm Demo 3 PRE: Circular Motion (2/2)

0 3 4

On a scale from 1-5, rate your confidence in your answer.



2pm Demo 3 POST: Circular Motion (1/3)

0 3 8

In this demo, an object is held at an angle from vertical. It is connected to the ceiling by a string. How should the instructor release the object such that it will travel in a circular path?

release the object from rest

0 %

push the object directly toward the center of the imaginary circle (toward vertical)

0 %

push the object directly away from the center of the imaginary circle (increasing the angle from vertical)

53 %

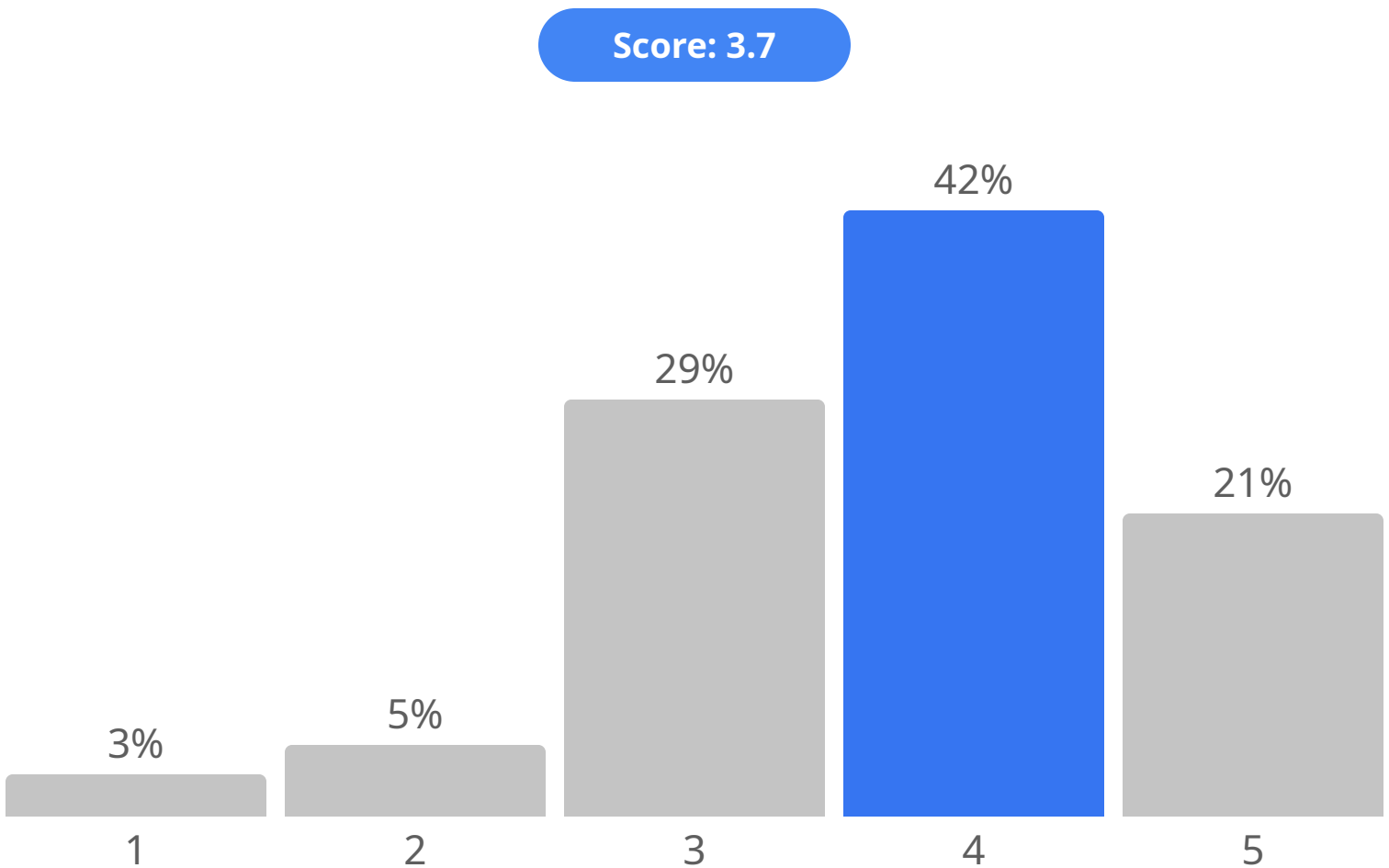
none of the above

47 %

2pm Demo 3 POST: Circular Motion (2/3)

0 3 8

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

2pm Demo 3 POST: Circular Motion (3/3)

0 3 7

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

19 %

Demonstration setup

11 %

Demonstration outcome

65 %

Physics equations

0 %

Something else

5 %

2pm Demo 4 PRE: Loop-de-Loop (1/2)

0 3 9

In this demo, a ball will roll down a ramp into a vertical loop-de-loop track. Modeling the motion of the ball completing the circular path, the required centripetal force will always:

Change the speed of the ball

5 %

Point toward the top of the loop

13 %

Be lesser in magnitude than the force of gravity

8 %

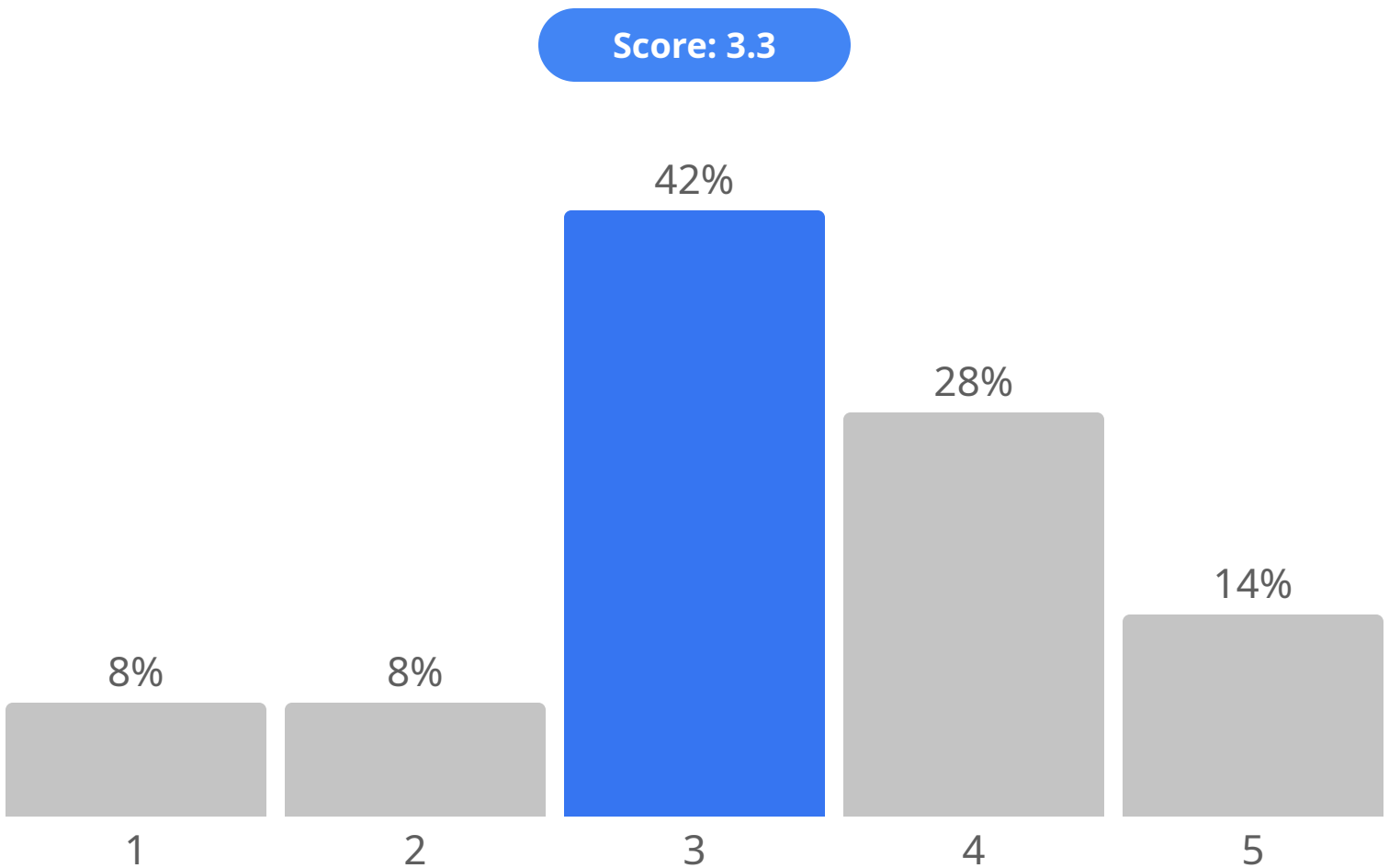
Point toward the center of the loop circle

74 %

2pm Demo 4 PRE: Loop-de-Loop (2/2)

0 3 6

On a scale from 1-5, rate your confidence in your answer.



2pm Demo 4 POST: Loop-de-Loop (1/3)

0 4 0

In this demo, a ball will roll down a ramp into a vertical loop-de-loop track. If the ball is to complete the circular path, the ramp must be:

Shorter than the circular loop

3 %

As tall as the circular loop

3 %

Taller than the circular loop

33 %

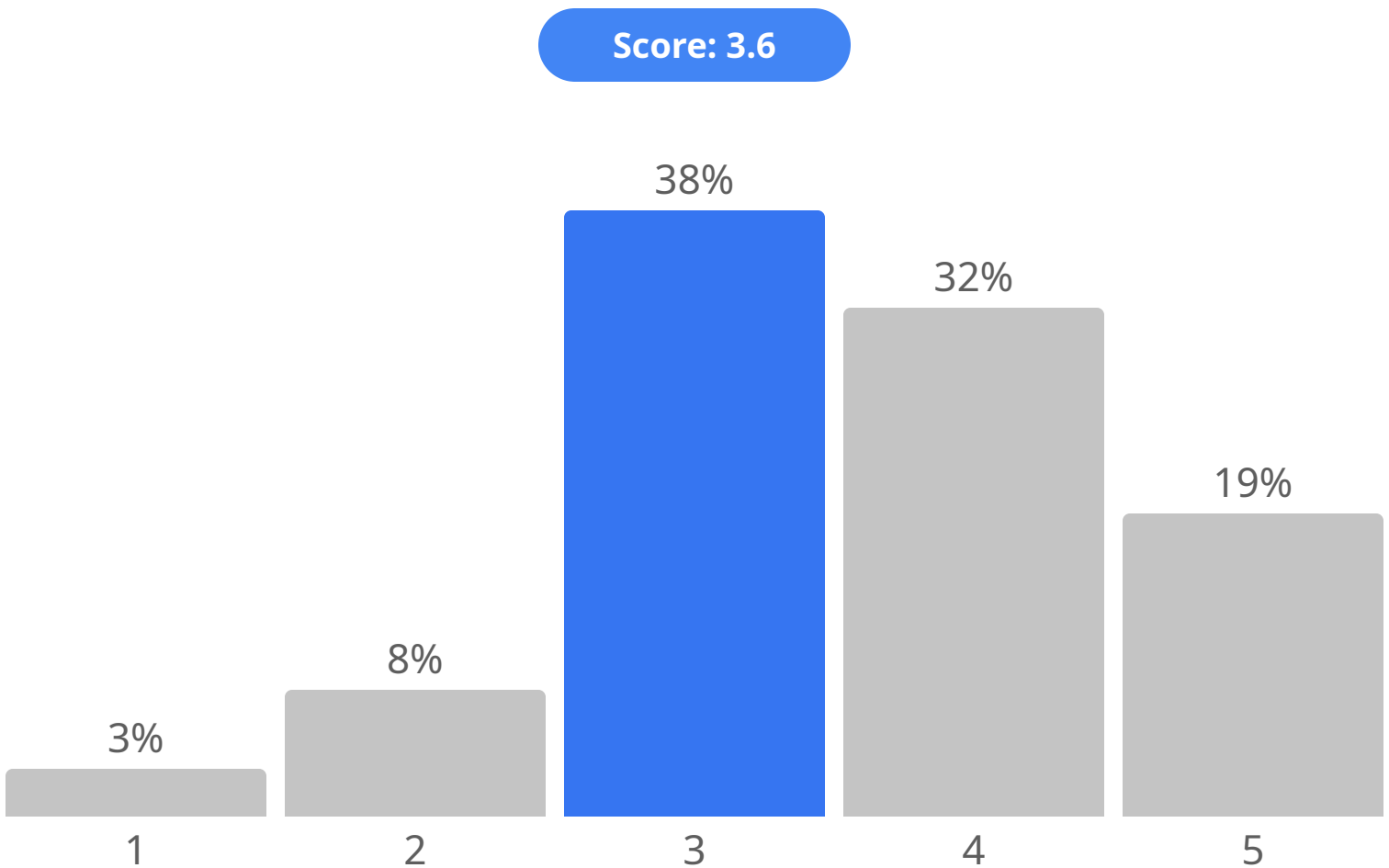
Two of these answers will work

63 %

2pm Demo 4 POST: Loop-de-Loop (2/3)

0 3 7

On a scale from 1-5, rate your confidence in your answer.

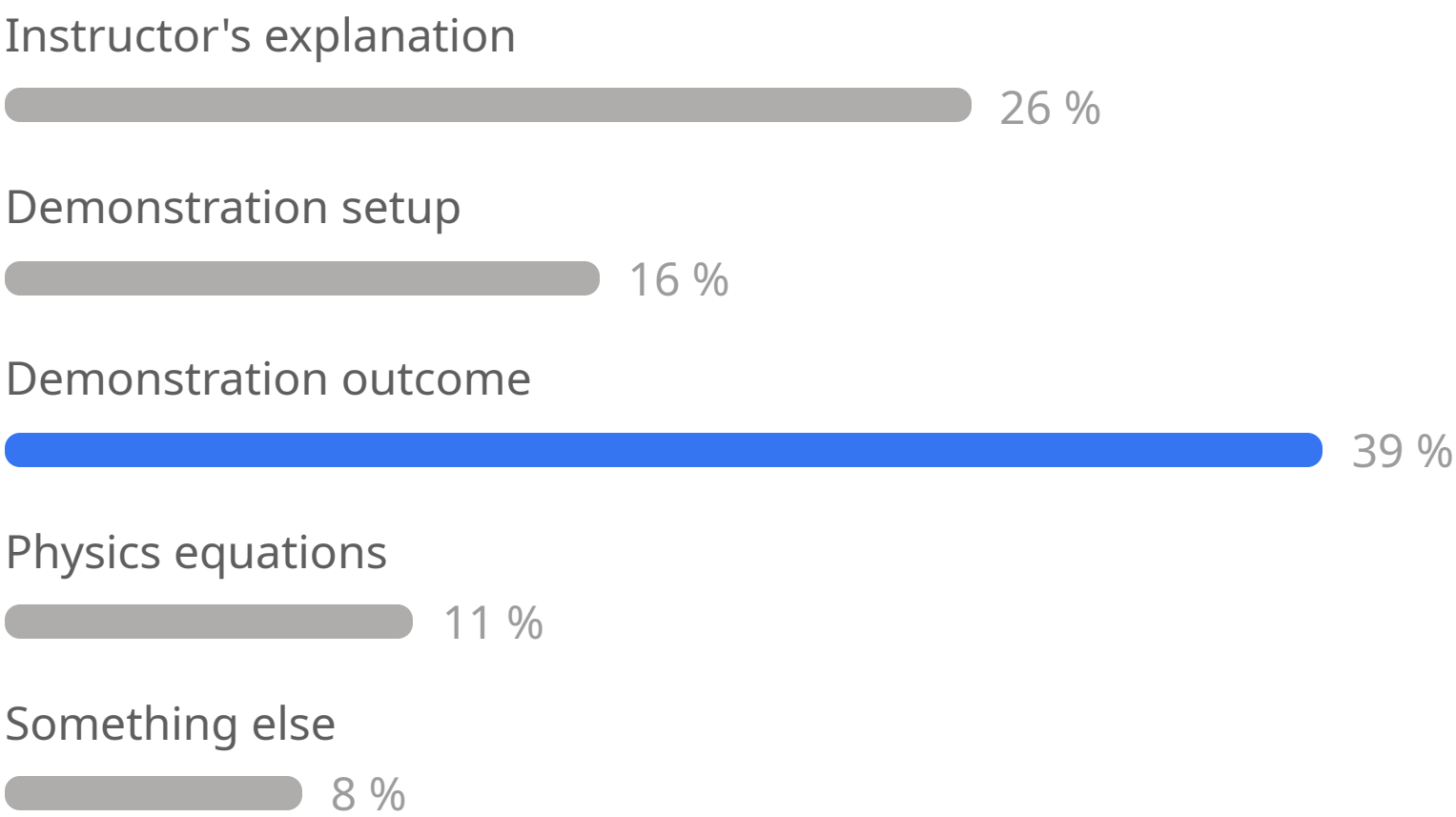


Multiple choice poll

2pm Demo 4 POST: Loop-de-Loop (3/3)

0 3 8

During the demonstration, which aspect were you focused on the most?



2pm Demo 5 PRE: Pendulum (1/2)

0 3 2

In this demo, a bowling ball is suspended from the ceiling via a rope, just like a pendulum. The instructor will hold the ball up to his chest and then release it. The instructor can safely:

Release the ball from rest

100 %

Push the ball forward

0 %

Both of these

0 %

Neither of these

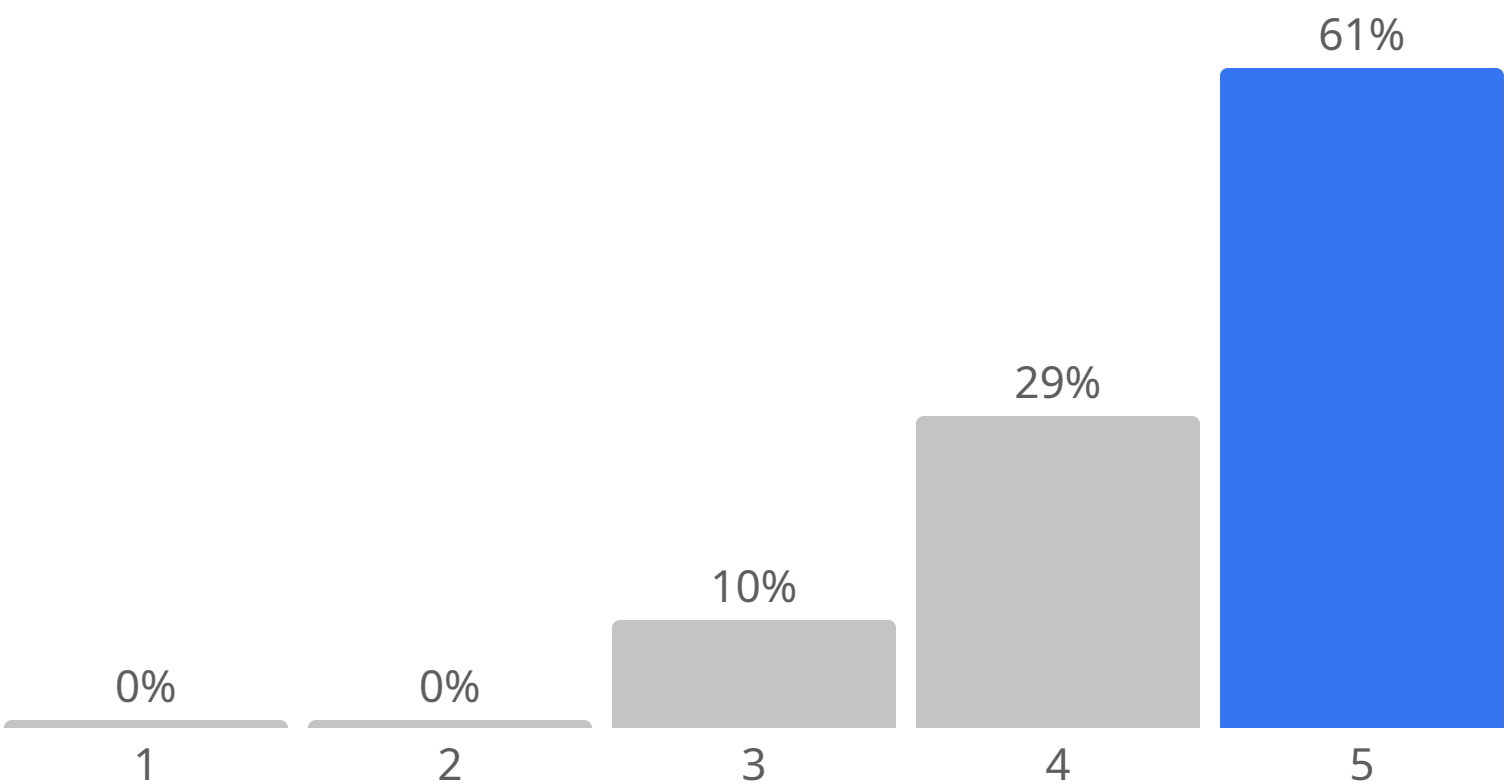
0 %

2pm Demo 5 PRE: Pendulum (2/2)

0 3 1

On a scale from 1-5, rate your confidence in your answer.

Score: 4.5



2pm Demo 5 POST: Pendulum (1/3)

0 3 1

In this demo, a bowling ball is suspended from the ceiling via a rope, just like a pendulum. The instructor will hold the ball up to his chest and then release it. Which of the following statements about the ball's mechanical energy is true:
(1/2)

The ball's potential energy is greatest at the lowest point of its trajectory.

0 %

If the energy is conserved, potential and kinetic energy will change by equal and opposite values.

94 %

The ball's kinetic energy is greatest at the highest point of its trajectory.

0 %

2pm Demo 5 POST: Pendulum (1/3)

0 3 1

In this demo, a bowling ball is suspended from the ceiling via a rope, just like a pendulum. The instructor will hold the ball up to his chest and then release it. Which of the following statements about the ball's mechanical energy is true:
(2/2)

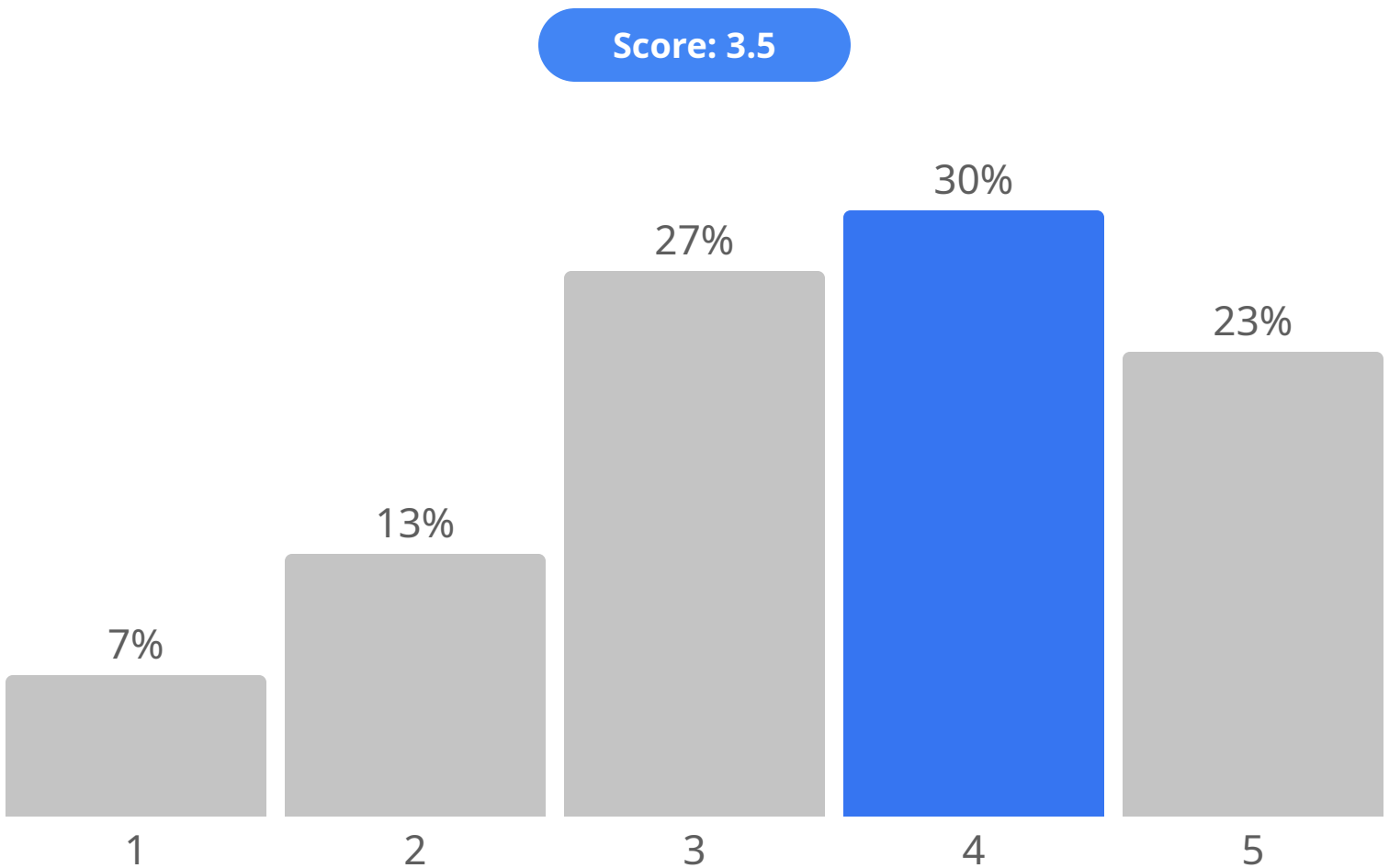
If the energy is conserved, the potential energy will be constant.

6 %

2pm Demo 5 POST: Pendulum (2/3)

0 3 0

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

2pm Demo 5 POST: Pendulum (3/3)

0 3 1

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

10 %

Demonstration setup

13 %

Demonstration outcome

71 %

Physics equations

3 %

Something else

3 %

Multiple choice poll

2pm Demo 6 PRE: Collisions (1/2)

0 3 6

Two balls of equal mass and different elasticity will roll down a ramp (from the same height). At the bottom of the ramp is an upright block of similar mass. Upon collision, which ball has a higher chance of toppling the block?

Happy (more elastic) ball

14 %

Sad (more inelastic) ball

83 %

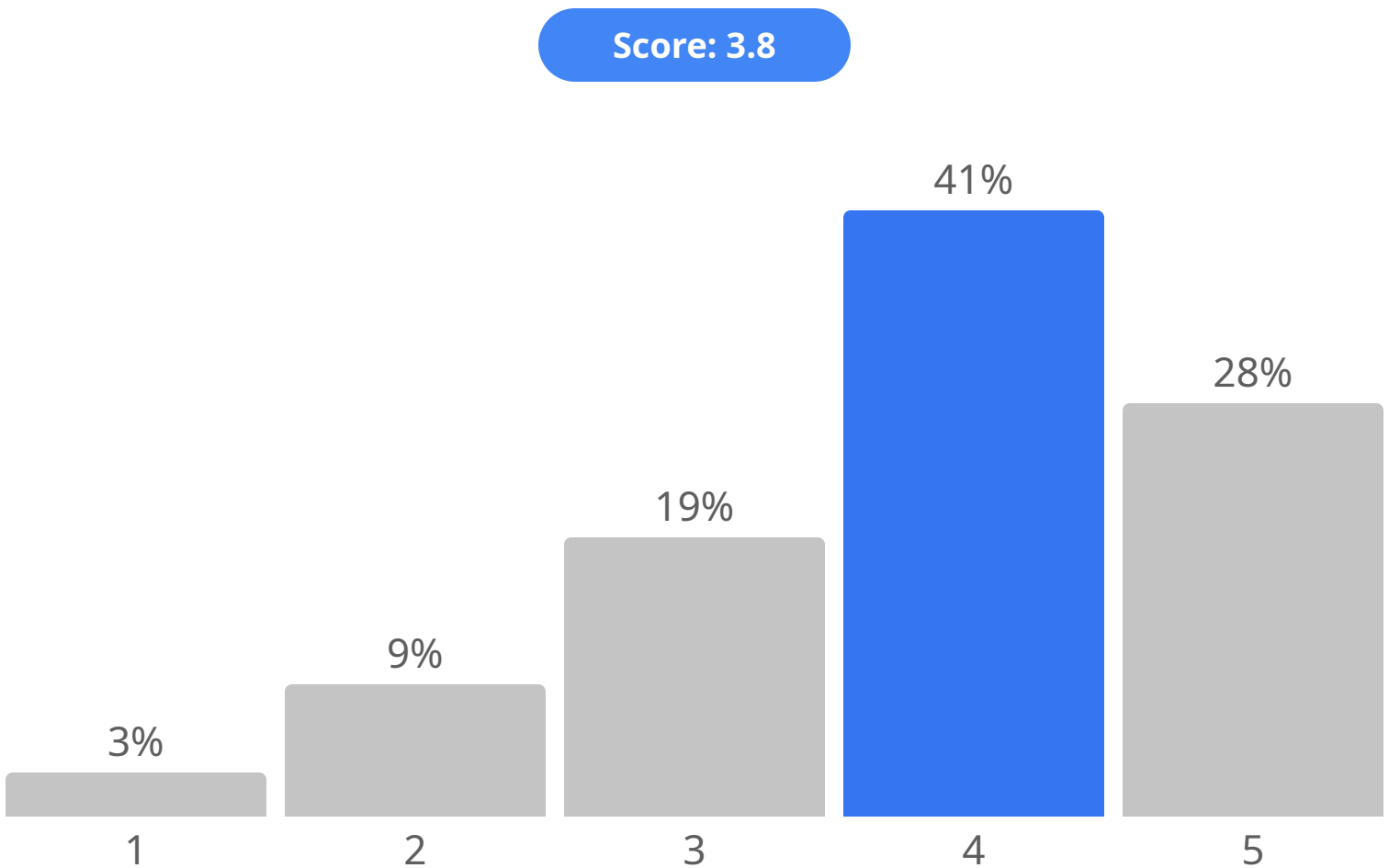
They have an equal chance

3 %

2pm Demo 6 PRE: Collisions (2/2)

0 3 2

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

2pm Demo 6 POST: Collisions (1/3)

0 3 6

This demo involves elastic and inelastic collisions. For inelastic collisions, what is conserved?

Total Kinetic Energy of the System

67 %

Total Momentum of the System

25 %

Both

3 %

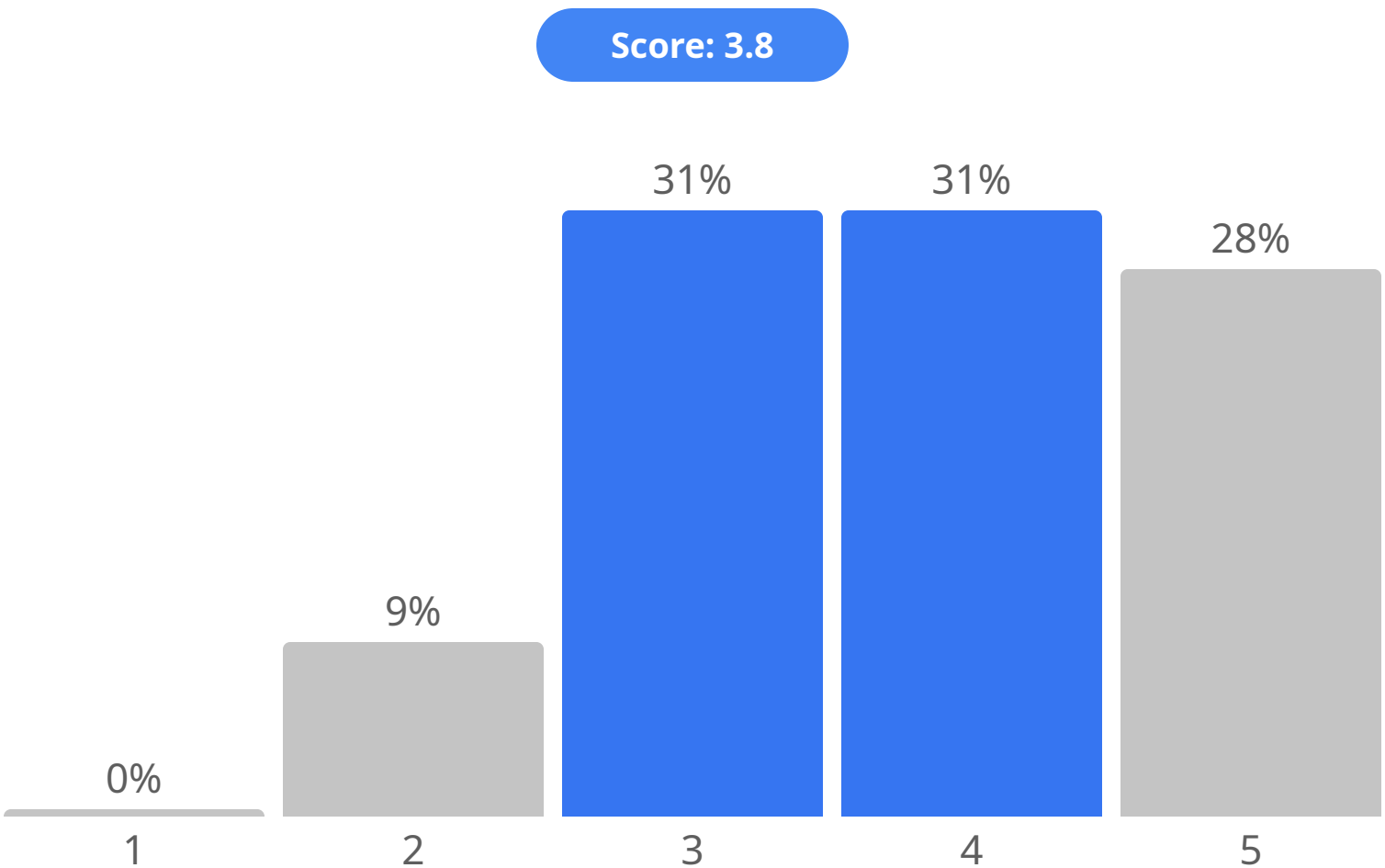
Neither

6 %

2pm Demo 6 POST: Collisions (2/3)

0 3 2

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

2pm Demo 6 POST: Collisions (3/3)

0 3 6

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

17 %

Demonstration setup

6 %

Demonstration outcome

69 %

Physics equations

3 %

Something else

6 %

2pm Demo 7 PRE: Rolling Down a Ramp (1/2)

0 3 1

In this demo, a solid sphere and a solid cylinder will roll down a ramp (released from rest at the same height). Which one will reach the bottom first?

They will reach the bottom at the same time

3 %

The sphere will reach the bottom first

74 %

The cylinder will reach the bottom first

6 %

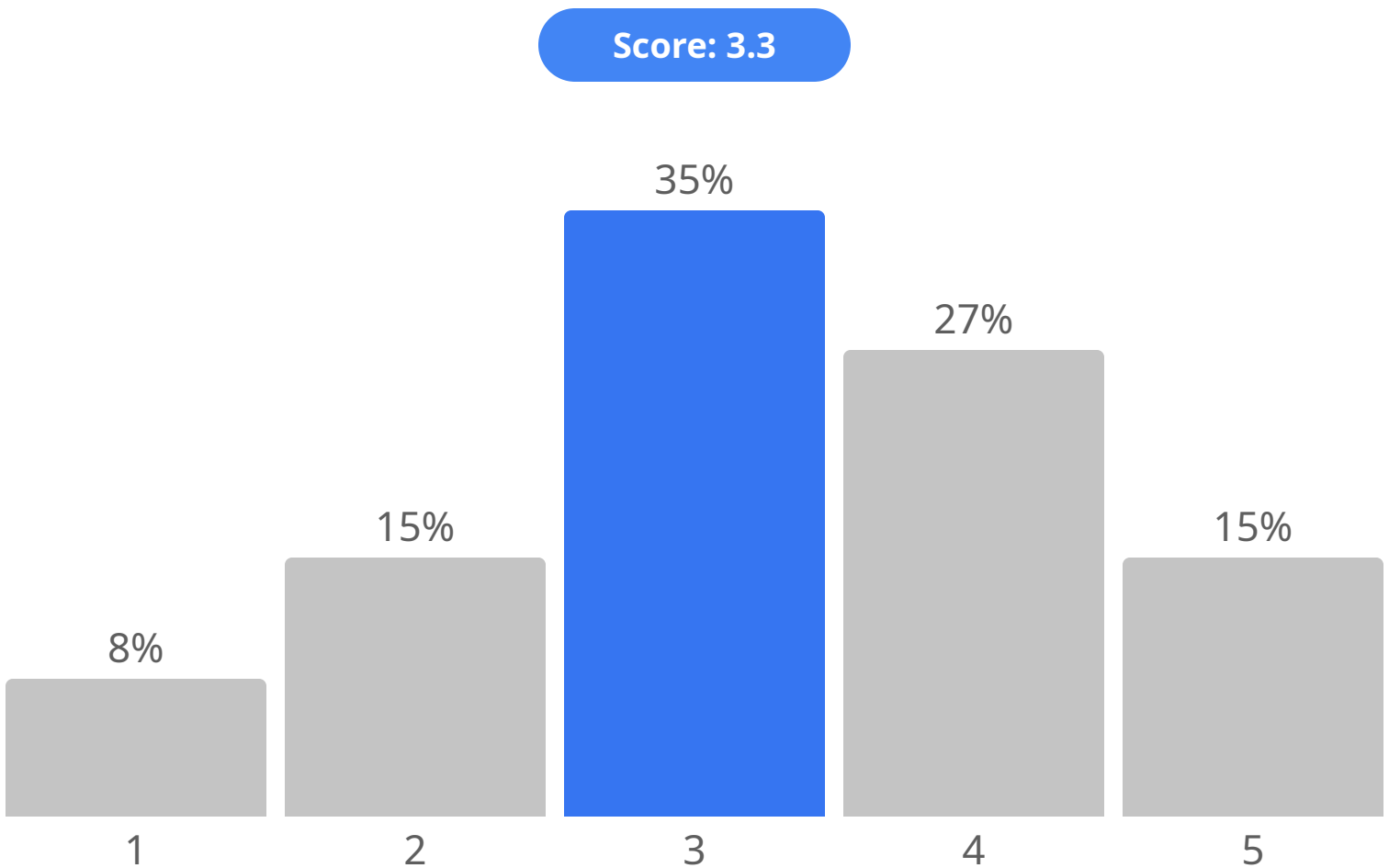
It will depend on the mass of each object

16 %

2pm Demo 7 PRE: Rolling Down a Ramp (2/2)

0 2 6

On a scale from 1-5, rate your confidence in your answer.



2pm Demo 7 POST: Rolling Down a Ramp (1/3)

0 2 9

In this demo, a sphere will roll down a ramp. Which of the following quantities will affect the time it takes the sphere to reach the bottom?

Mass of the sphere

7 %

Radius of the sphere

3 %

Both

3 %

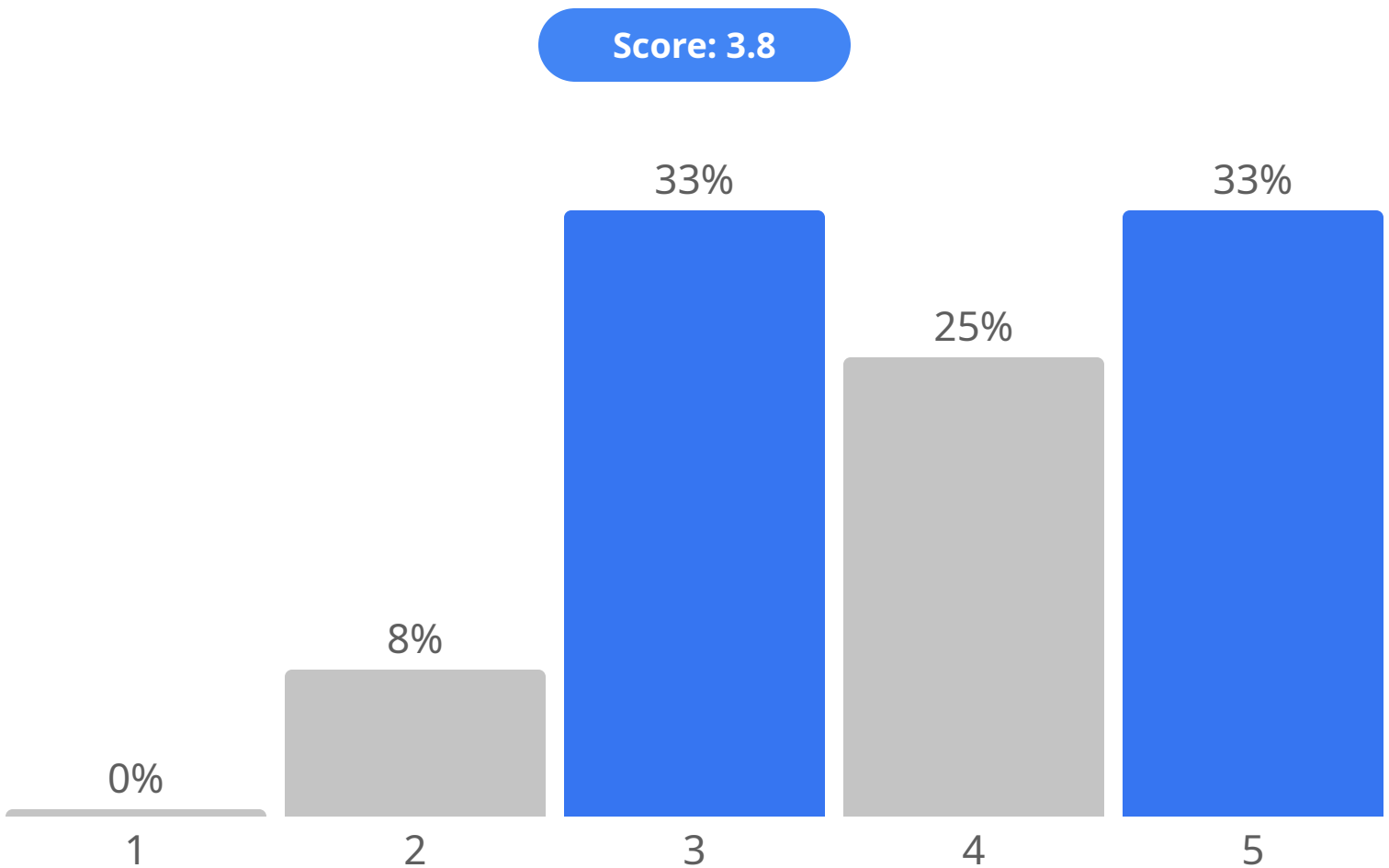
Neither

86 %

2pm Demo 7 POST: Rolling Down a Ramp (2/3)

0 2 4

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

2pm Demo 7 POST: Rolling Down a Ramp (3/3)

0 2 9

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

0 %

Demonstration setup

10 %

Demonstration outcome

83 %

Physics equations

3 %

Something else

3 %

2pm Demo 8 PRE: Dumbbell Torque (1/2)

0 3 2

In this demo, a participant sits on a swiveling chair while holding a dumbbell in each hand. They begin with their arms out wide and then bring the dumbbells into their chest. Upon doing this, their angular velocity will:

Increase

100 %

Decrease

0 %

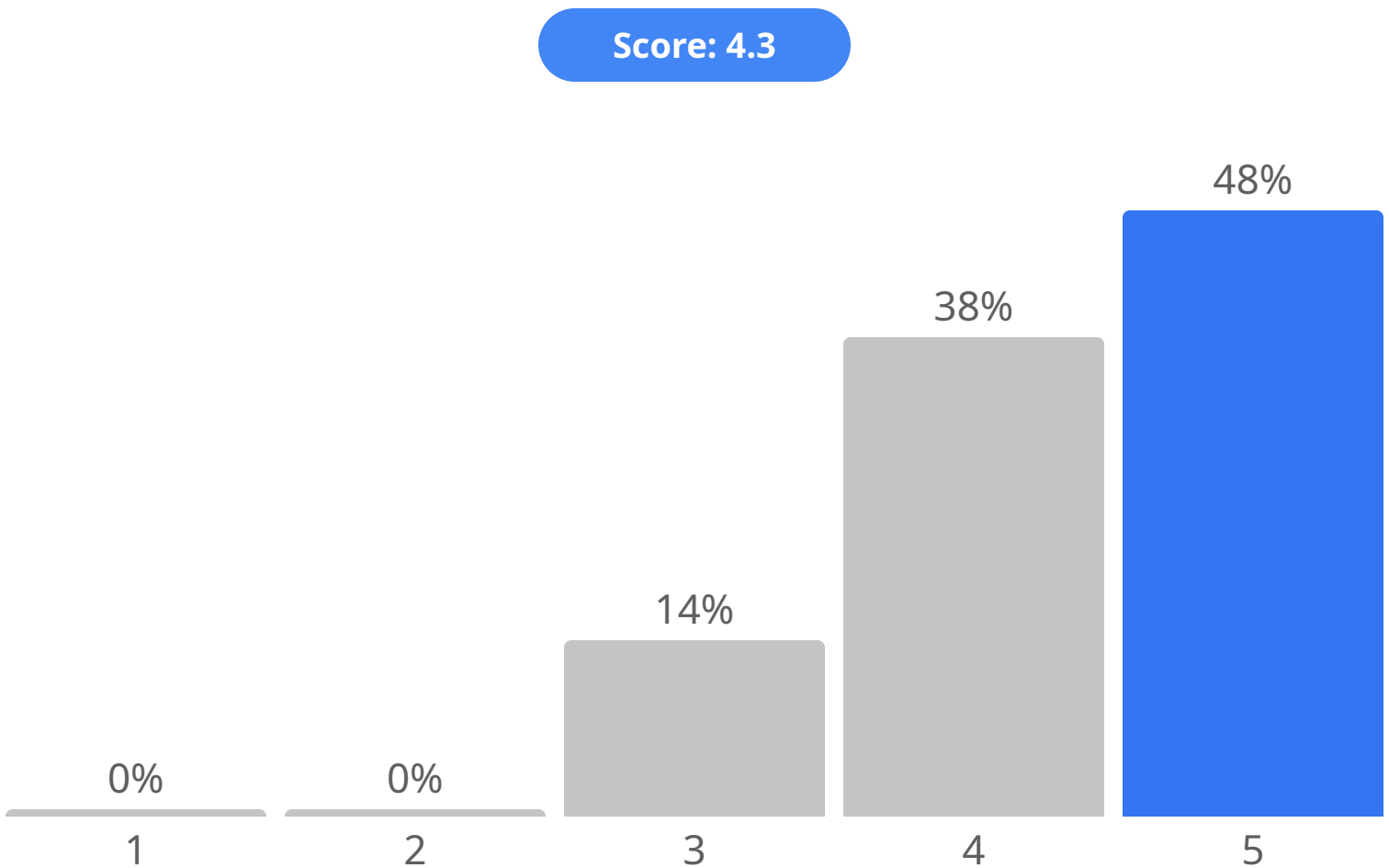
Stay the same

0 %

2pm Demo 8 PRE: Dumbbell Torque (2/2)

0 2 9

On a scale from 1-5, rate your confidence in your answer.



2pm Demo 8 POST: Dumbbell Torque (1/3)

0 3 4

In this demo, a participant sits on a swiveling chair while holding a dumbbell in each hand. They begin with their arms out wide and then bring the dumbbells into their chest. Which of the following is NOT true:

The participant exerts a torque

6 %

The participant causes an angular acceleration on the system

74 %

The participant preserves the kinetic energy of the system

18 %

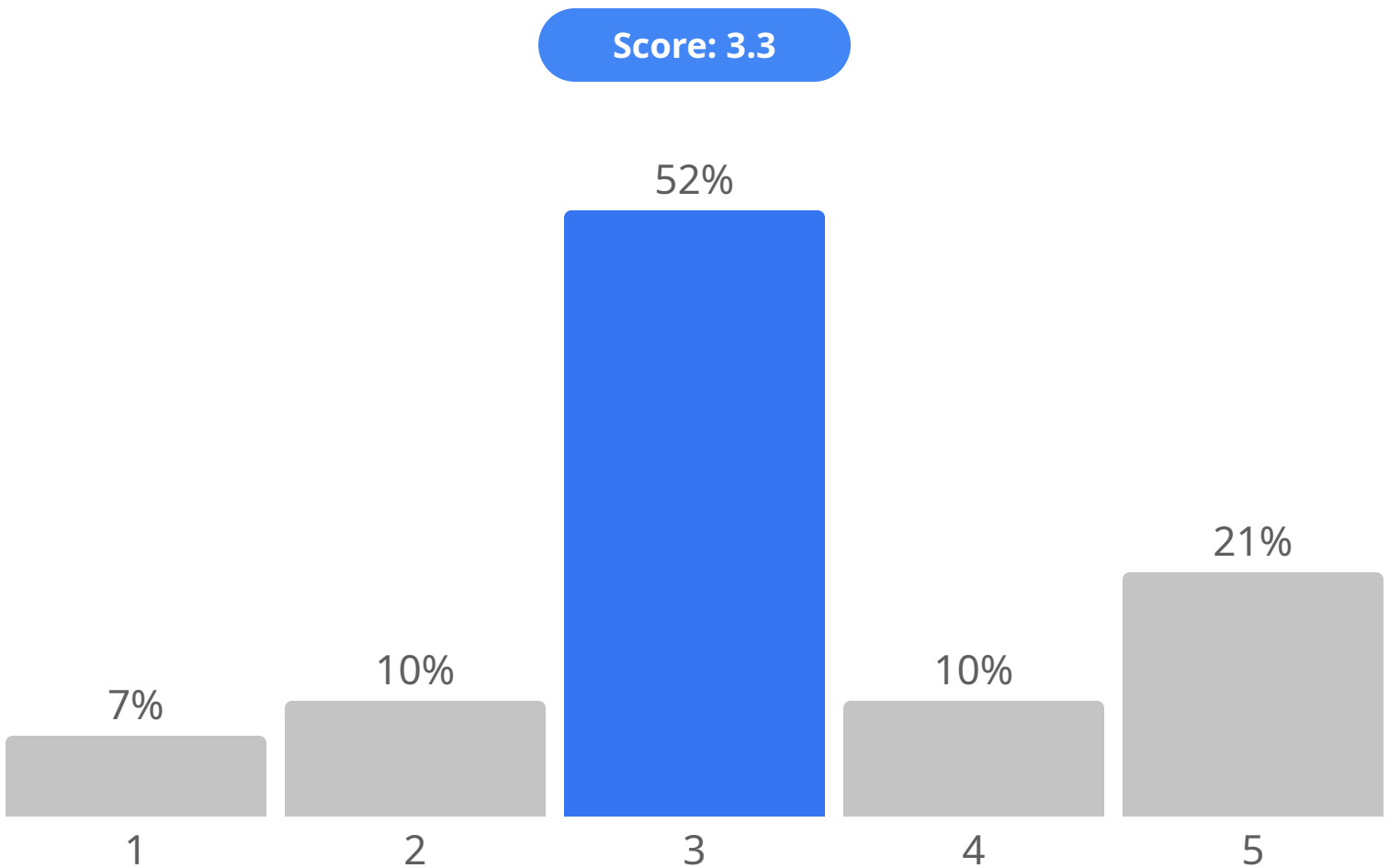
The participant changes the rotational inertia of the system

3 %

2pm Demo 8 POST: Dumbbell Torque (2/3)

0 2 9

On a scale from 1-5, rate your confidence in your answer.



Multiple choice poll

2pm Demo 8 POST: Dumbbell Torque (3/3)

0 3 4

During the demonstration, which aspect were you focused on the most?

Instructor's explanation

6 %

Demonstration setup

18 %

Demonstration outcome

74 %

Physics equations

0 %

Something else

3 %